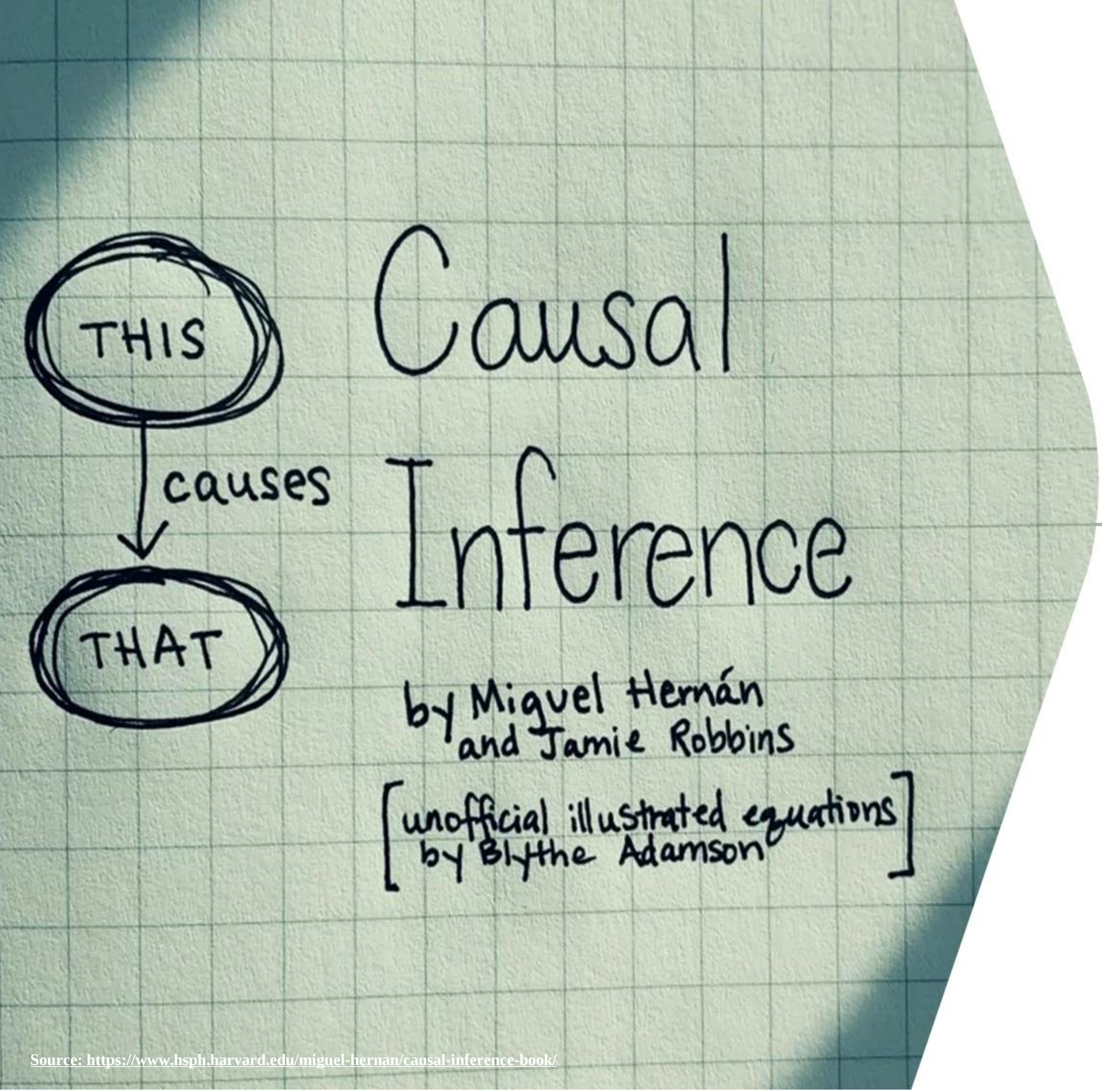

Source: https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/

한국인공지능학회 동계 단기강좌

Potential Outcome and Directed Acyclic Graph Approaches to Causality

Jiyong Park

Bryan School of Business and Economics

University of North Carolina at Greensboro

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What is Causal Inference?

A Science of Cause and Effect

What is Causality?

Identifying Cause and Effect

KIDS ACADEMY

An effect is WHAT happens.
A cause is WHY something happens.

Check the best way to finish each sentence.

Jimmy was happy

☐ because he won the marathon.

☐ because he lost his keys.

https://www.kidsacademy.mobi/printables/indentifying-cause-and-effect-worksheets/

What is Causality?

Maggie's T-shirt
was dirty

- [ ] because she ate chocolate candies.
- [ ] because she watched too much TV.

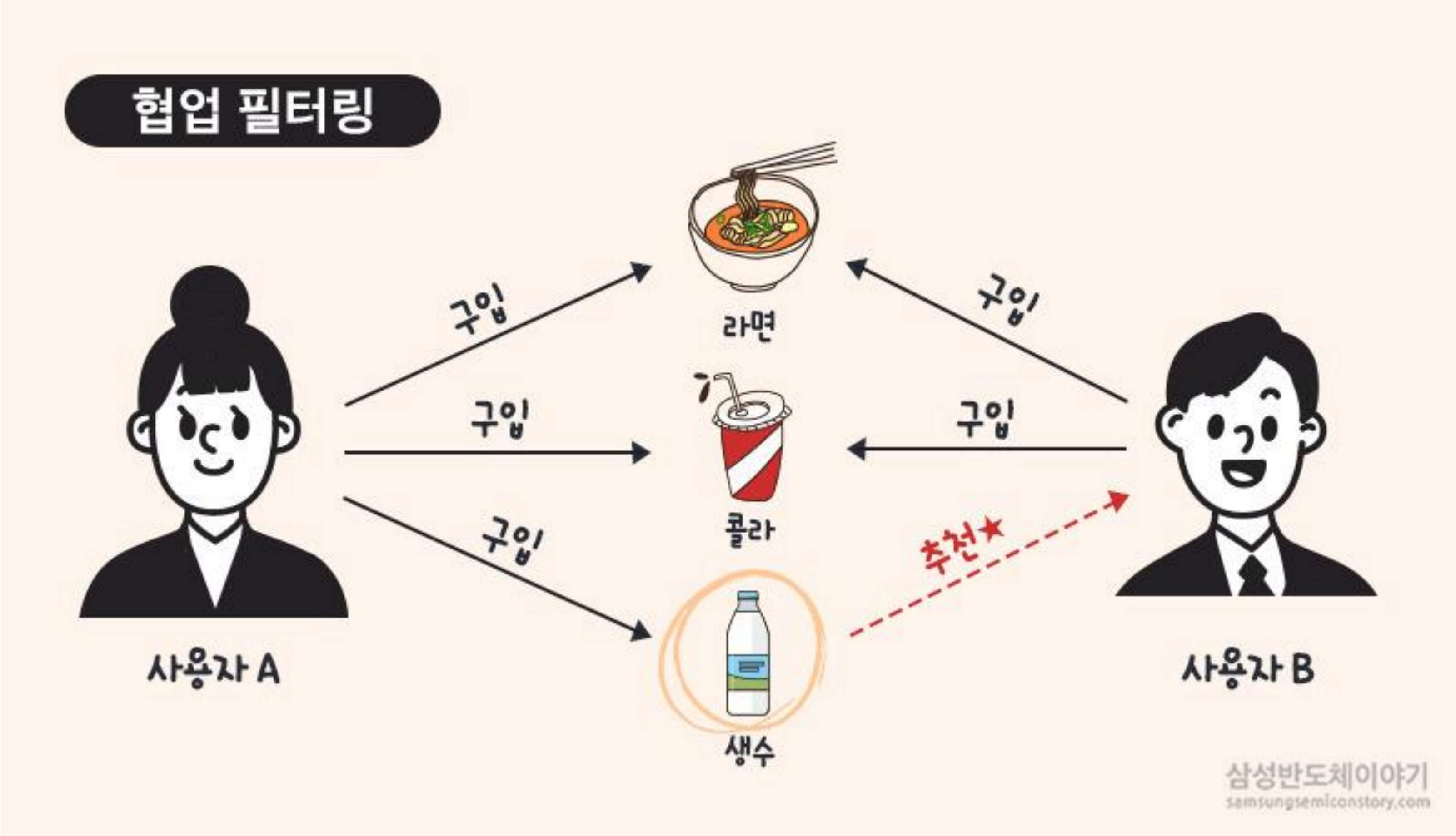
Emma felt sick

- [ ] so Mom took her to the doctor.
- [ ] so she went to the zoo.

https://www.kidsacademy.mobi/printables/indentifying-cause-and-effect-worksheets/

Motivating Examples (1) Recommender Systems

- Does the recommender system *cause* the increase in product sales?



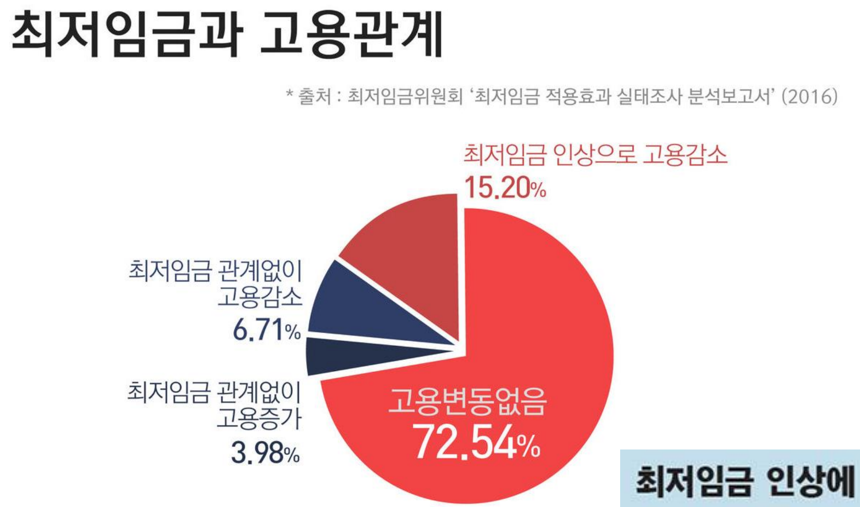
https://www.samsungsemiconstory.com/

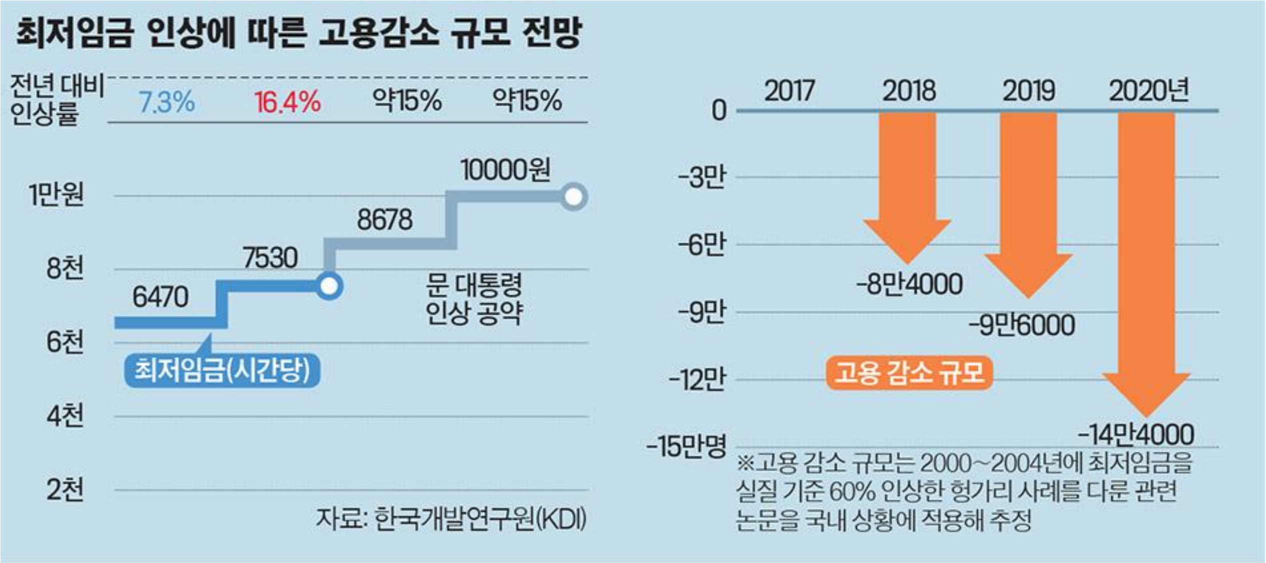
Why did Users A and B purchase Ramen and Coke?

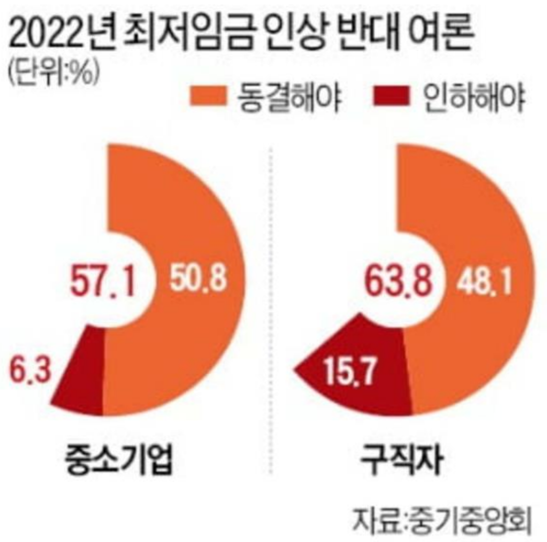
Are User B likely to purchase Water even without the recommendation?

Motivating Examples (2) Minimum Wage and Employment

- Does the increase in minimum wage *cause* the decrease in employment?





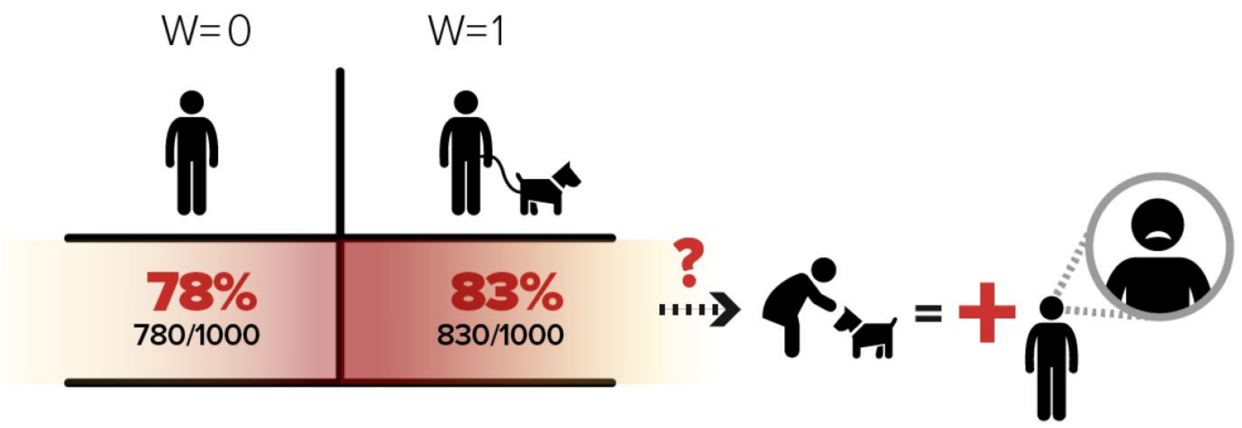


https://news.sbs.co.kr/news/endPage.do?news_id=N1003669103
https://jmagazine.joins.com/economist/view/321648
https://www.hankyung.com/economy/article/2021061658691

Motivating Examples (3) Companion Animals and Depression

- Does keeping companion animals *cause* the decrease in depression?

https://magazine.hankyung.com/money/article/202101203871c



Depression rate

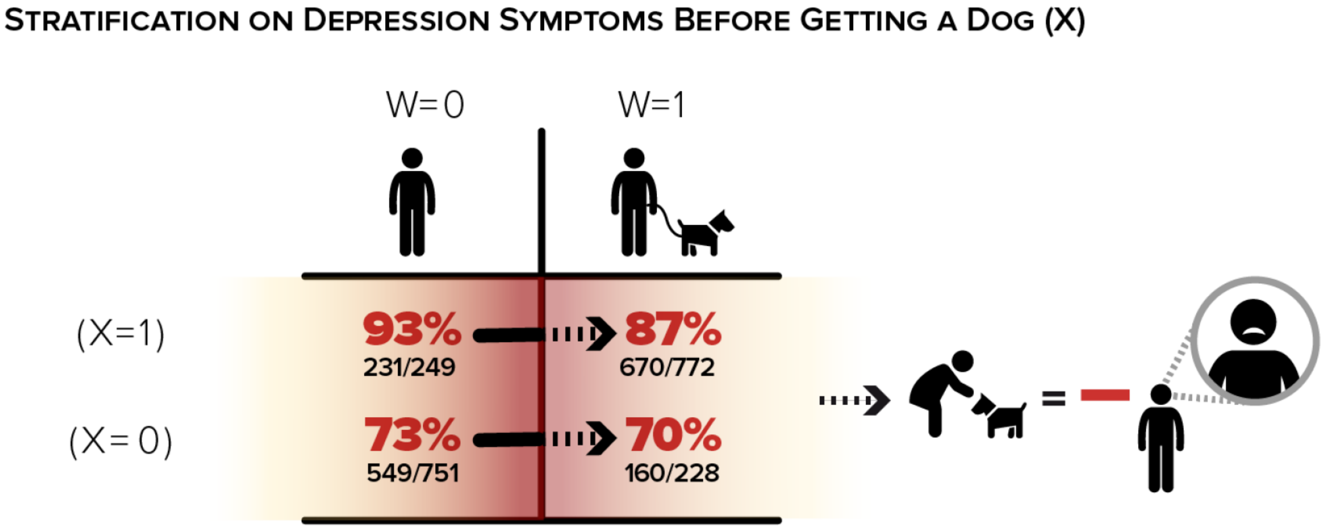
Does keeping companion animals *increase* depression rates?

Dominici, F., Bargagli-Stoffi, F.J. and Mealli, F., 2020. From controlled to undisciplined data: estimating causal effects in the era of data science using a potential outcome framework. *arXiv preprint arXiv:2012.06865*.

Motivating Examples (3) Companion Animals and Depression

- Does keeping companion animals *cause* the decrease in depression?

https://magazine.hankyung.com/money/article/202101203871c



Depression rate

People with a higher risk of depression tend to adopt a companion animal.

Dominici, F., Bargagli-Stoffi, F.J. and Mealli, F., 2020. From controlled to undisciplined data: estimating causal effects in the era of data science using a potential outcome framework. *arXiv preprint arXiv:2012.06865*.

Motivating Examples (4) Medical Treatment

- Which treatment is more effective to decrease mortality rates?

Simpson's Paradox

Mortality rate	Mild Symptom	Severe Symptom	Total
Treatment A	15% (195/1300)	30% (30/100)	**16.1% (225/1400)**
Treatment B	**10% (10/100)**	**20% (100/500)**	18.3% (110/600)

$P(Y|T, S = mild)$ $P(Y|T, S = severe)$ $P(Y|T)$

It depends on the causal structure between treatment, symptom, and outcome.

What is Causal Inference?

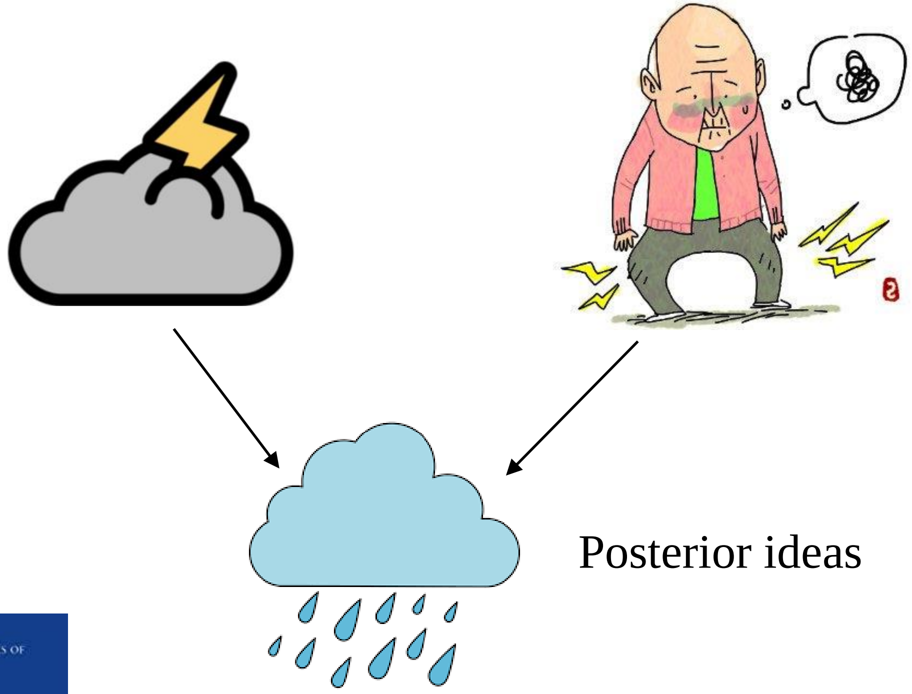
Causal inference encompasses statistical and computational methods for studying such questions using data and determining what causes what.

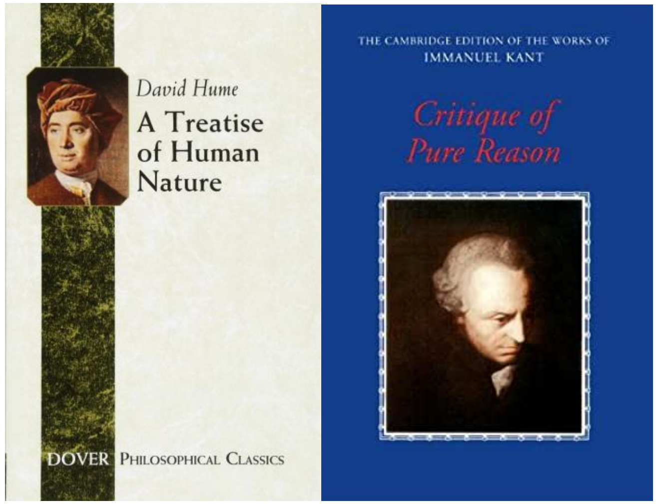
What is Causality?

- Arguments between David Hume and Immanuel Kant

"An object precedent and contiguous to another, and so united with it, that the idea of the one determined the mind to form the idea of the other, and the impression of the one to form a more lively idea of the other."

"Causality does not occur in reality. It is a mental construction due to the repeated and constant conjunction between the event we call cause and that we call effect."





"I freely admit that it was the remembrance of David Hume which, many years ago, first interrupted my dogmatic slumber and gave my investigations in the field of speculative philosophy a completely different direction."

What is Causality?

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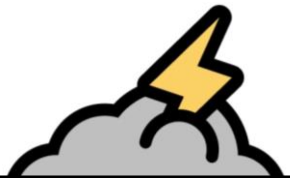
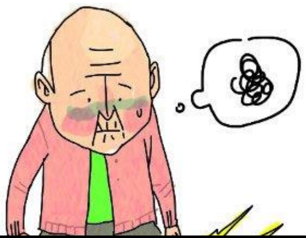


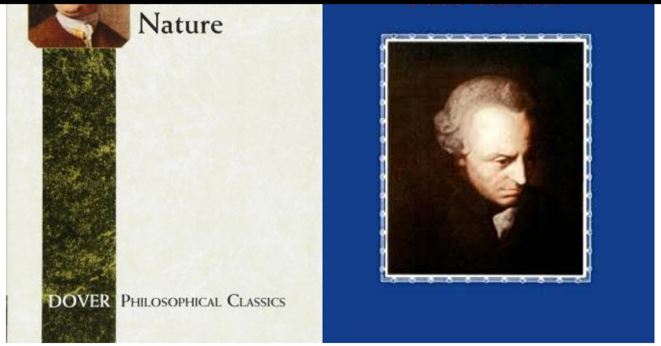


For data scientists and analysts,

a more systematic, scientific–rather than philosophical–approach to

causality is needed.

mental construction due to the repeated and constant conjunction between the event we call cause and that we call effect.”



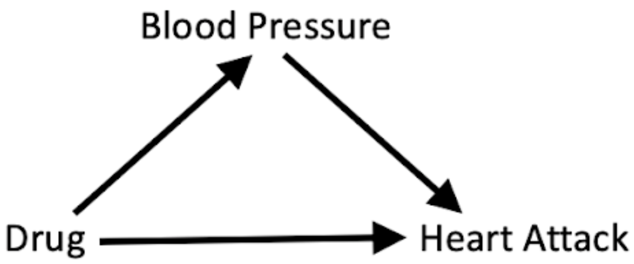
“I freely admit that it was the remembrance of David Hume which, many years ago, first interrupted my dogmatic slumber and gave my investigations in the field of speculative philosophy a completely different direction.”

Causal Inference Frameworks

Potential Outcomes Framework (Rubin Causal Model)	Structural Causal Model (Pearl Causal Model)

Table 1.1. ***Example of Potential Outcomes and Causal Effect with One Unit***

Unit	Potential Outcomes		Causal Effect
	Y(Aspirin)	Y(No Aspirin)	
You	No Headache	Headache	Improvement due to Aspirin



Causal Inference Frameworks

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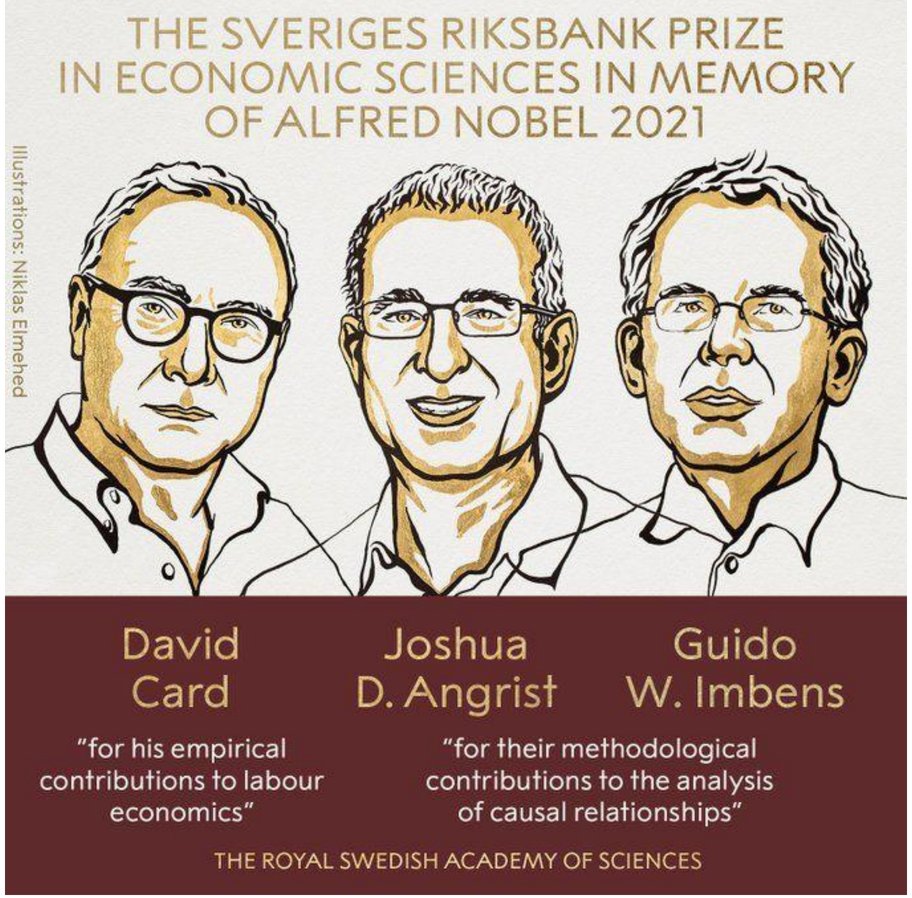


ACM A. M. Turing Award

Judea Pearl

United States – 2011

Citation:

For fundamental contributions to artificial intelligence through the development of a calculus for probabilistic and causal reasoning.

ML/AI without Causal Models Reply on Correlation

What if..? 인과추론의 데이터과학

Most ML models (without any causal models) achieve high accuracy by taking advantage, in an automated manner, of all correlations that exist in the data to predict what label or outcome is most associated with a given set of features (or variables).

Hünermund, P., Kaminski, J. and Schmitt, C., 2021. Causal Machine Learning and Business-Decision Making. *Working Paper* (https://causalscience.org/Working-Paper.pdf)

FIGURE 1: Classification of popular data analysis algorithms.

	Prediction	Causal Inference: Observational	Causal Inference: Experimental
Statistics/Econometrics	ANOVA	Difference-in-Differences	A/B Testing
	Linear Regression	Instrumental Variables	Business Experimentation
	Logistic Regression	Propensity Score Matching	Randomized Controlled Trials
	Time Series Forecasting	Regression Discontinuity	
Machine Learning	Boosting	Additive Noise Models	Causal Reinforcement Learning
	Decision Trees & Random Forests	Causal Forests	Multiarmend Bandits
	Lasso, Ridge & Elastic Net	Causal Structure Learning	Reinforcement Learning
	Neural Networks	Directed Acyclic Graphs	
	Support Vector Machines	Double/Debiased Machine Learning	

Note: Examples of popular data analysis algorithms in statistics and econometrics, as well as machine learning and artificial intelligence, classified according to prediction and causal inference methods. Causal inference methods are further differentiated according to observational (based on ex-post observed data) and experimental approaches.

Big Data Might Help, but Cannot Solve Causal Inference

Can Big Data Solve the Fundamental Problem of Causal Inference?

Rocío Titiunik, *University of Michigan*

BIG DATA AS LARGE N *(number of samples)*

"Large n does not automatically remove or even alleviate what is often the most challenging step of empirical research… An estimator that is inconsistent remains inconsistent regardless of how large the number of observations used."

BIG DATA AS LARGE P *(number of variables)*

"This idea is promising indeed because it implies that big data as large p eventually will allow us to solve automatically the fundamental problem of causal inference by "controlling for" massive amounts of information using sophisticated algorithms, computers, and statistical assumptions."

"How do we know if our large p dataset contains all relevant controls and excludes all post-treatment variables?"

Titiunik, R., 2015. Can Big Data Solve the Fundamental Problem of Causal Inference?. *PS: Political Science & Politics*, 48(1), pp.75-79.

Two Paradigms of Data Science / Analytics

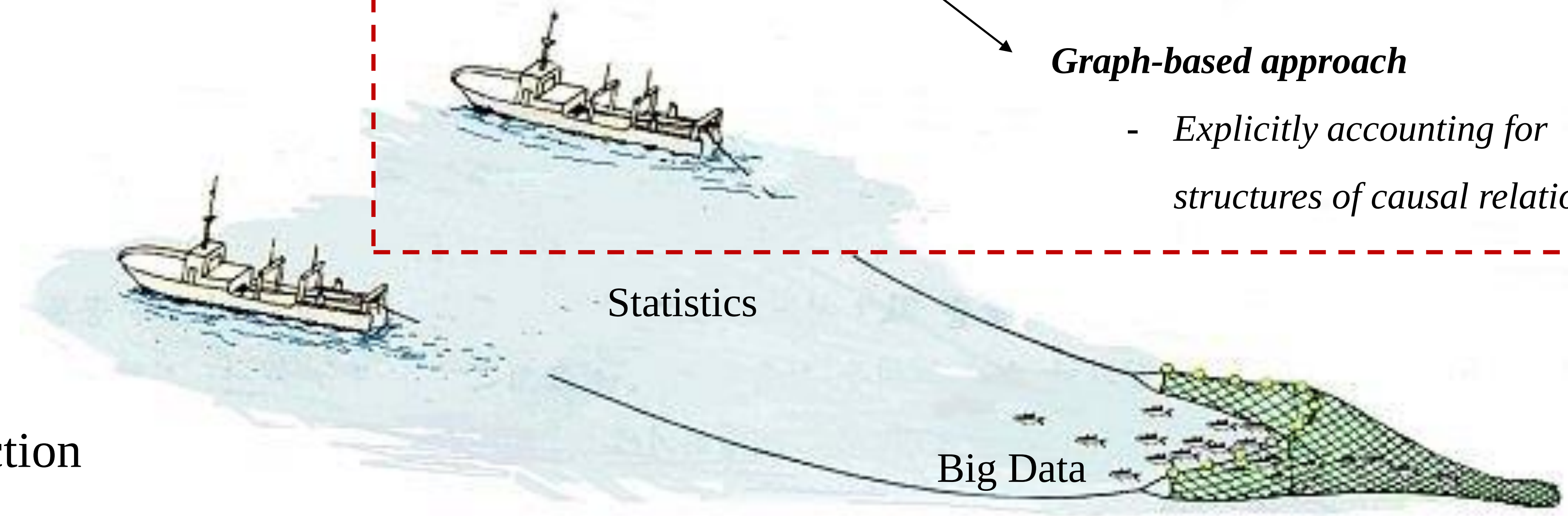
Causal Inference

Design-based approach

- *Removing selection bias by design*

Graph-based approach

- *Explicitly accounting for structures of causal relationships*



Statistics

Big Data

Prediction

- *Model Fitting Approach (e.g., Machine Learning)*
- *Information Filtering Approach (e.g., Recommendation Systems)*

Data Science / Analytics

Further Learning

- [인과추론의 어려움]
- [[Session 1-1] 인과추론의 다양한 접근법]
- [[Session 11-1] 인과추론과 예측 방법론의 차이]

Potential Outcome Framework

A Design-Based Approach

Potential Outcome Framework

- Causation is defined as the difference in potential outcomes after the treatment.
    - *"What if the treatment was not applied?"*
    - Causal effect of the treatment = (Actual outcome for treated if treated) – (Potential outcome for treated if not treated)

Potential Outcome Framework

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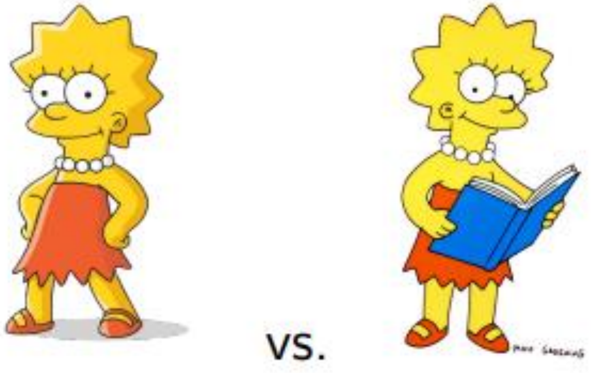
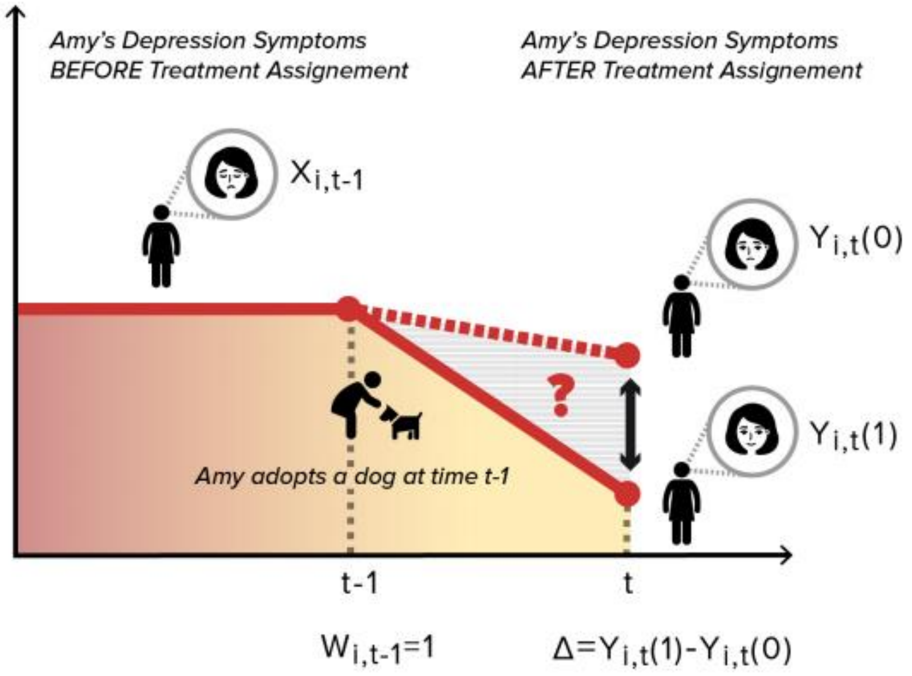
Counterfactual

Example

Causal effect of reading on grades



Counterfactual



Dominici, F., Bargagli-Stoffi, F.J. and Mealli, F., 2020. From controlled to undisciplined data: estimating causal effects in the era of data science using a potential outcome framework. *arXiv preprint arXiv:2012.06865*.

Fundamental Problem of Causal Inference

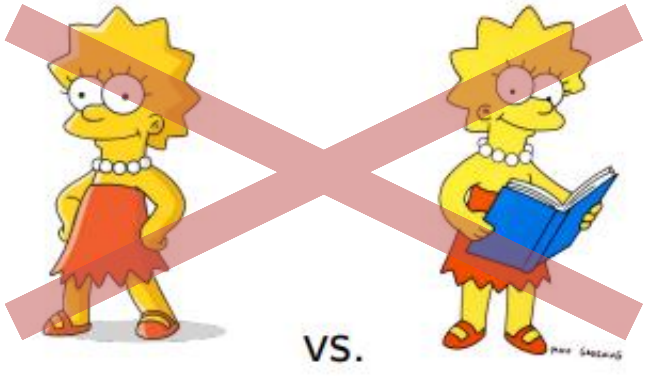
- By definition, potential outcomes are not always observable. The fundamental challenge in causal inference is that we do not observe both potential outcomes; we only observe one.
  - Causal effect of the treatment = (Actual outcome for treated if treated) – (Potential outcome for treated if not treated)

    Counterfactual
  - But, in reality, we can only observe:

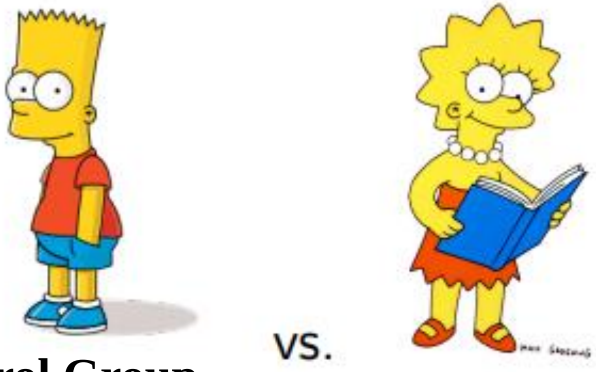
    = (Actual outcome for treated if treated) – (Actual outcome for untreated if not treated)

    Control Group

Estimate we can actually obtain



Counterfactual



Control Group

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Counterfactual

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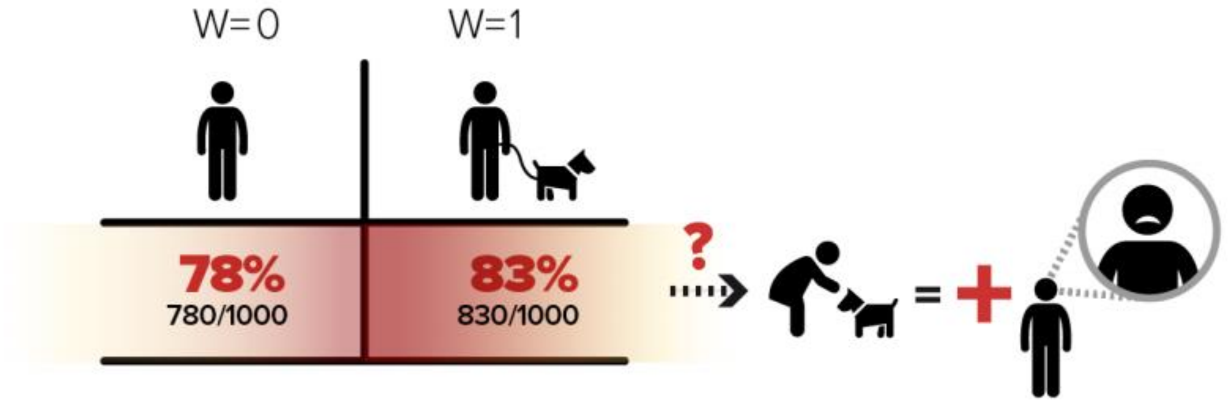
  - But, in reality, we can only observe:

Control Group

  = (Actual outcome for treated if treated) – (Actual outcome for untreated if not treated)

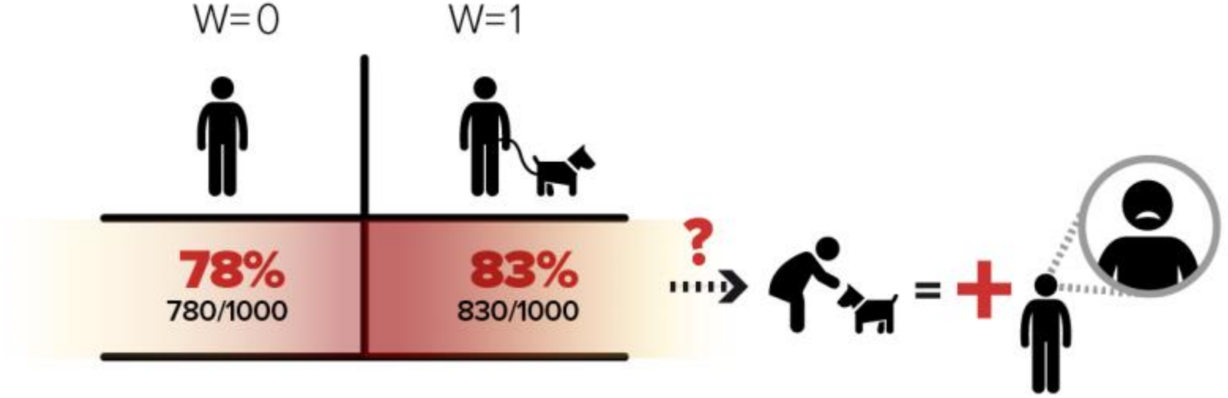


Dominici, F., Bargagli-Stoffi, F.J. and Mealli, F., 2020. From controlled to undisciplined data: estimating causal effects in the era of data science using a potential outcome framework. *arXiv preprint arXiv:2012.06865*.

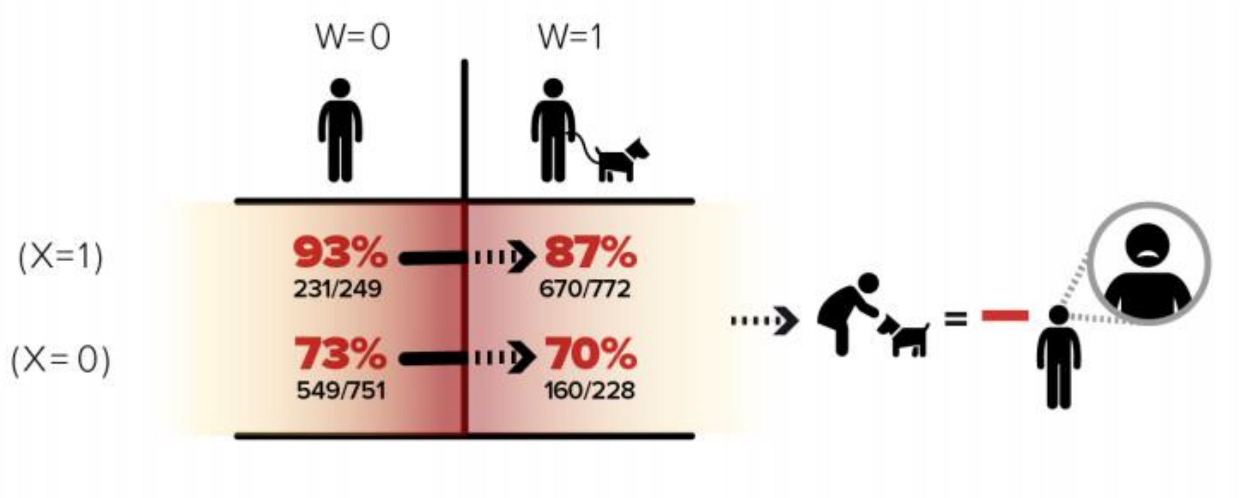
Selection Bias

- In reality, treatments are not assigned randomly. Individuals select the treatment (but we don't know why).
- As a result, the treatment and control groups may be systematically different and not be comparable.

Causal effect of adopting a dog?



Are they comparable, except the adoption?



Dominici, F., Bargagli-Stoffi, F.J. and Mealli, F., 2020. From controlled to undisciplined data: estimating causal effects in the era of data science using a potential outcome framework. *arXiv preprint arXiv:2012.06865*.

Selection Bias

- Selection bias is the systematic difference between the treatment group in the absence of treatment (counterfactual) and the control group.

- Decomposition of causal effect and selection bias
    - **Observed effect of the treatment** = (Outcome for treated if treated) – (Outcome for untreated if not treated)

= (Outcome for treated if treated)

Zero sum

– (Outcome for treated if not treated) + (Outcome for treated if not treated)

– (Outcome for untreated if not treated)

Causal effect

= (Outcome for treated if treated) – (Outcome for treated if not treated)

+ (Outcome for treated if not treated) – (Outcome for untreated if not treated)

Selection bias

= **Causal effect + Selection bias**

Causal Inference ≅ Finding Comparable Control Group

- From the perspective of potential outcome framework, causal inference is to remove the selection bias.

  = *(Outcome for treated if not treated)* – *(Outcome for untreated if not treated)*

- How? the treatment group should be comparable to the control (untreated) group in the absence of treatment.
- Causal inference is to take advantage of research designs in which the control groups are comparable to the treatment groups in all aspects, on average, but the fact that they are treated or not.

The most important principle of causal inference

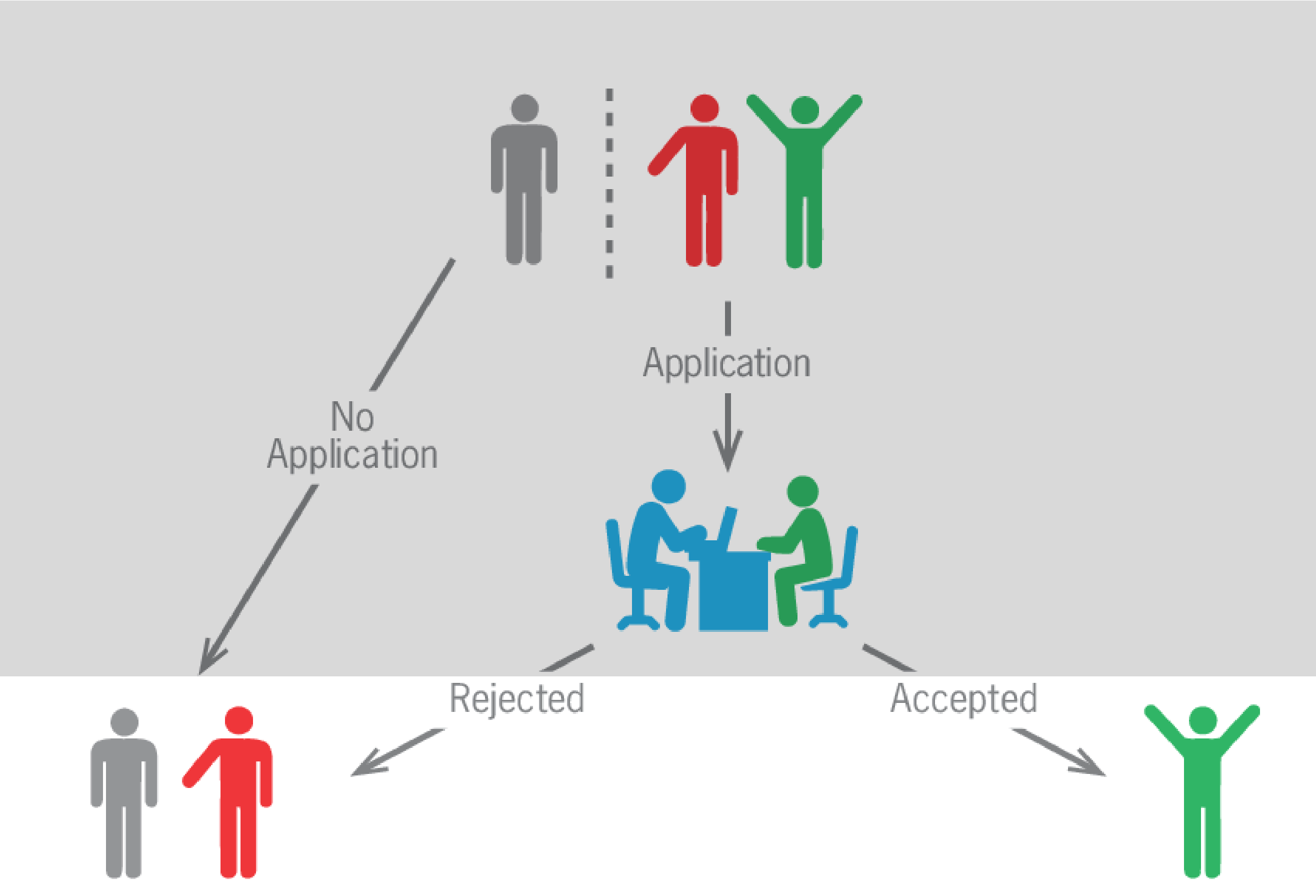
from the perspective of potential outcome framework

Treatment Group w/o Treatment (Counterfactual)

Control Group

Taking Selection Bias Out by Design

- Selection bias is unavoidable in most observational studies.



Control Group without Grant

Treatment Group with Grant

In most observation studies, a selection process (data-generation process) is unobservable.

Causal effect of grant?

Taking Selection Bias Out by Design

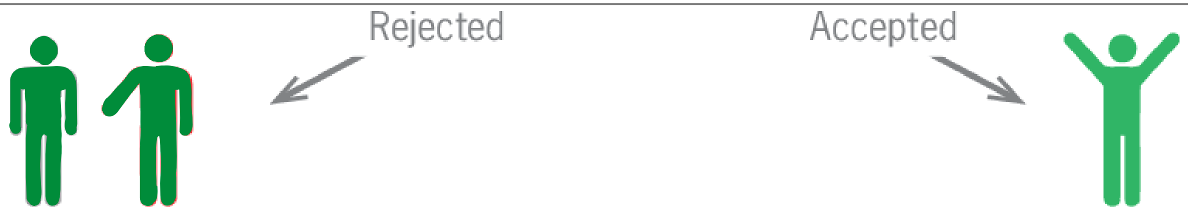
- Selection bias is unavoidable in most observational studies. But research designs can help mitigate it.

1. Research Designs for Causal Inference

   (≅ Treatment Assignment Mechanisms)

   a. Randomized Controlled Trial
   b. (Natural) Quasi-Experiment
   c. Instrumental Variable

2. Causal Graphs (Graphical Modeling)
3. Selection Models (Statistical Modeling)



Control Group without Grant

Treatment Group with Grant

Causal effect of grant?

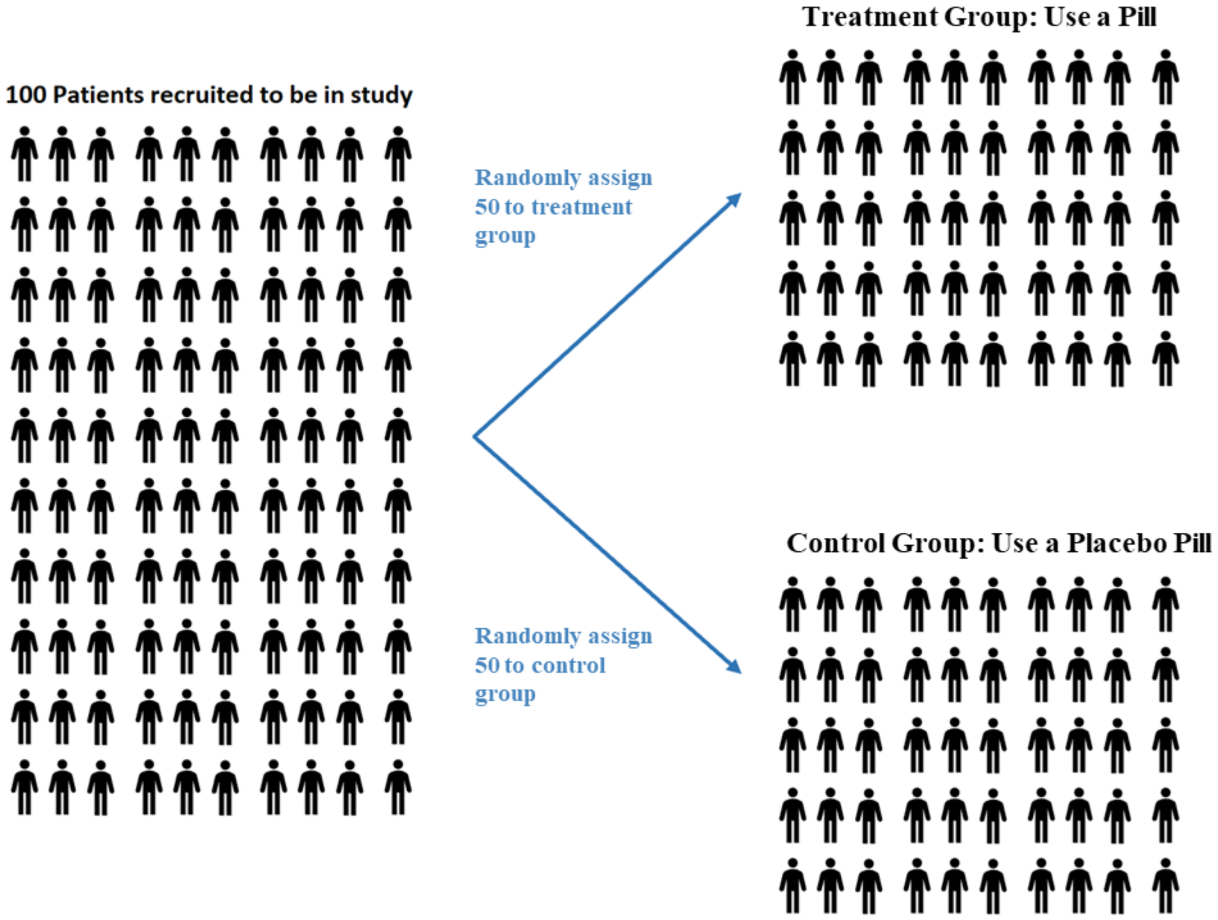
Gold Standard of Causal Inference: Random Assignment

- Relying on the law of large numbers, random assignment ensures that the subjects with various characteristics are distributed evenly across the treatment and control groups, so that they are comparable.

Gold Standard of Causal Inference: Random Assignment

- Relying on the law of large numbers, random assignment ensures that the subjects with various characteristics are distributed evenly across the treatment and control groups, so that they are comparable.

Male : Female = 0.5 : 0.5
w/ kids : w/o kids = 0.4 : 0.6
Average age = 32
Average heights = 170 cm
Average blood pressure = 100
etc…



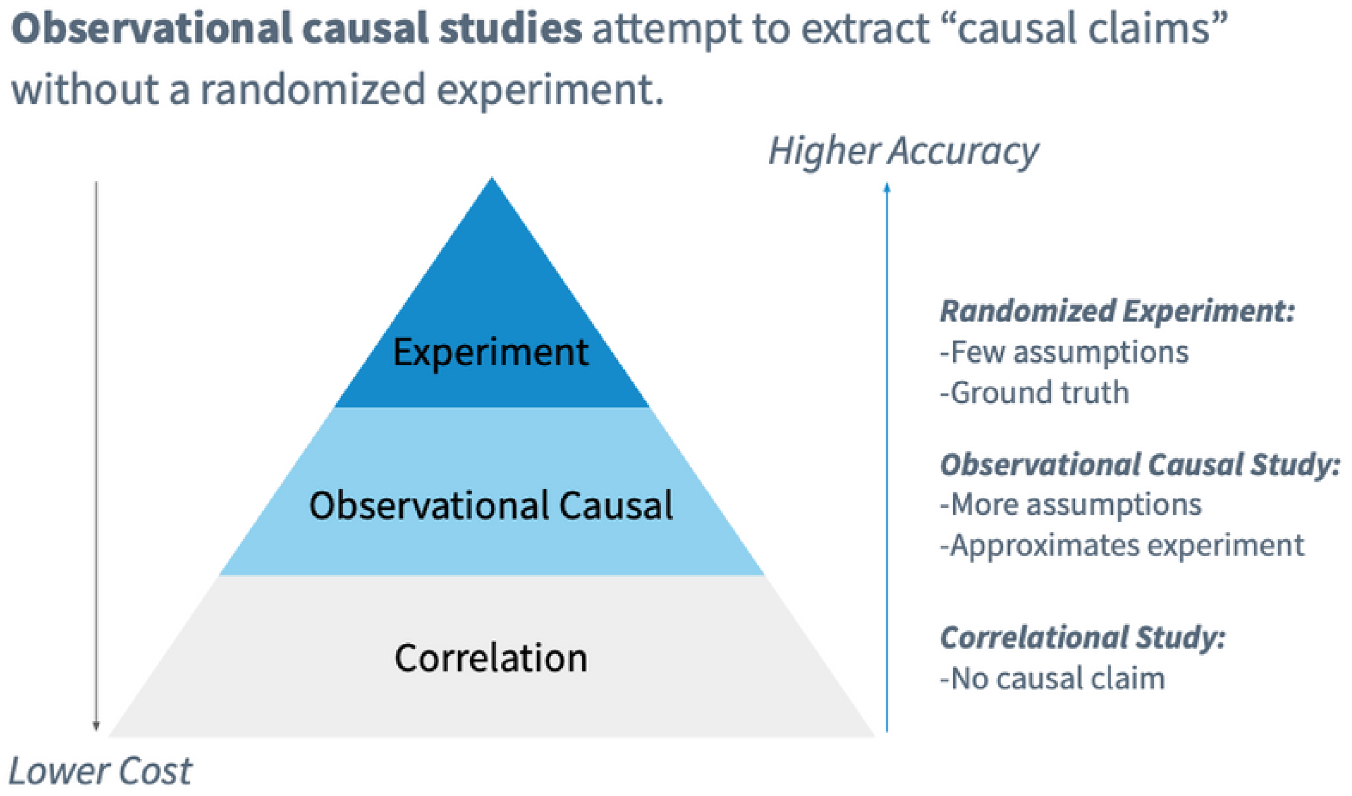
Male : Female = 0.5 : 0.5
w/ kids : w/o kids = 0.4 : 0.6
Average age = 31.5
Average heights = 169 cm
Average blood pressure = 95
etc…

Male : Female = 0.5 : 0.5
w/ kids : w/o kids = 0.4 : 0.6
Average age = 32
Average heights = 171 cm
Average blood pressure = 105
etc…

Random Assignment is not Always Feasible

- Quasi-experiments aim to mimic the randomized experiment by assigning the treatment and control groups based on an exogenous shock, self-selection, or an arbitrary cutoff, instead of random assignment.

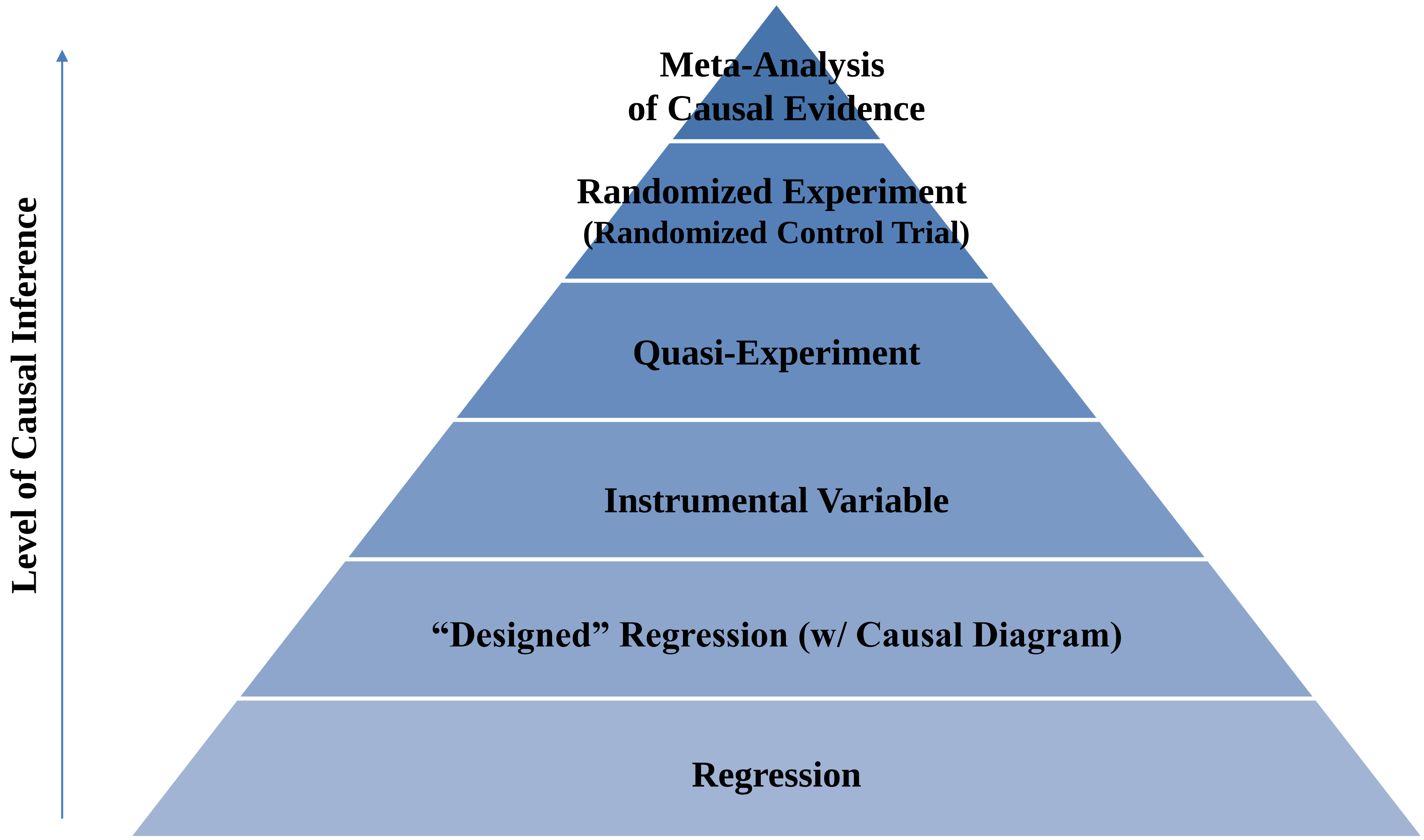
- The goal of quasi-experiments is to seek the comparable control group and infer the counterfactual from the control group.



Bojinov, I., Chen, A. and Liu, M., 2020. The importance of being causal. Harvard Data Science Review 2.3.

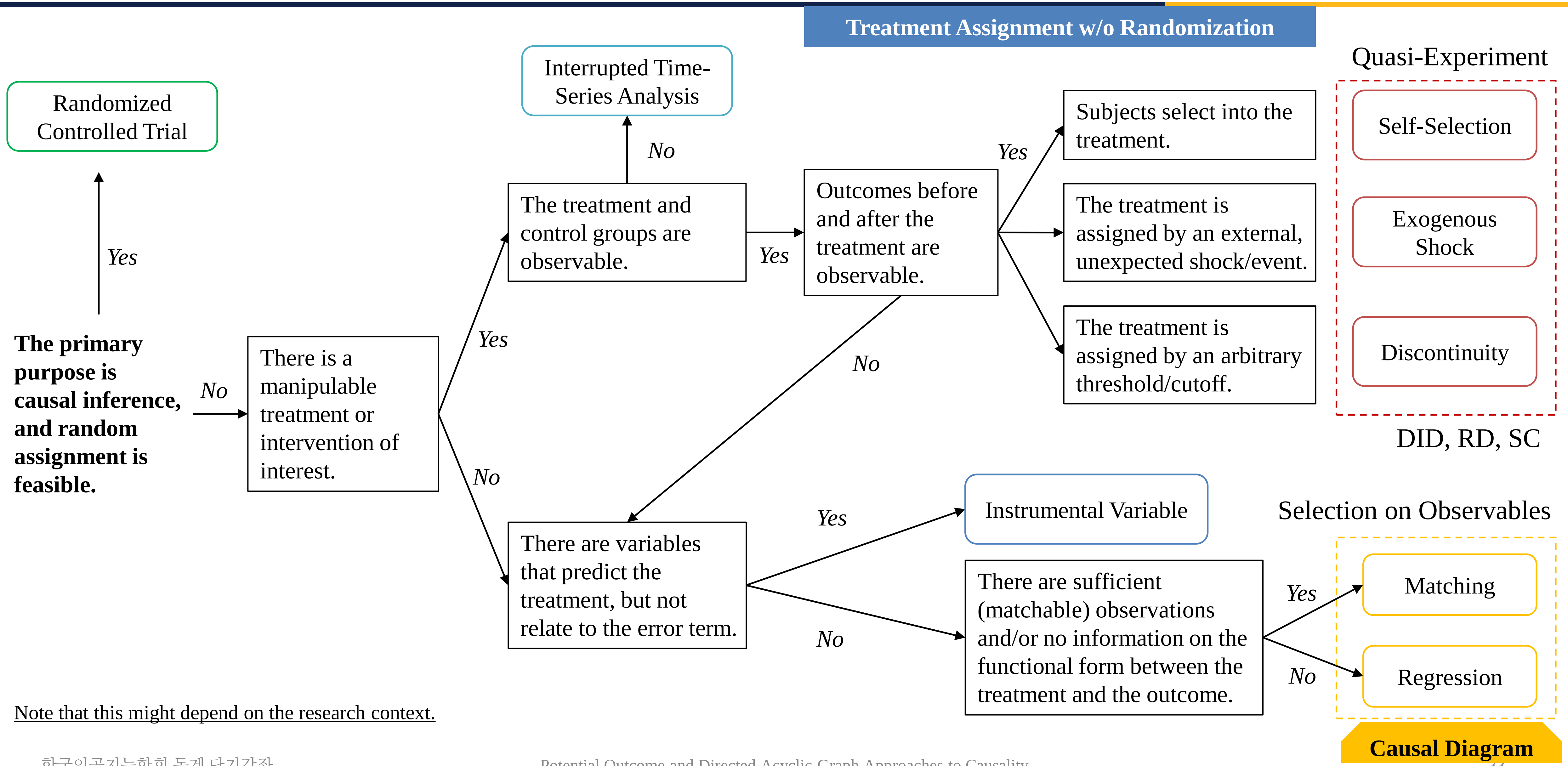
Causal Hierarchy of Research Design for Causal Inference

인과추론의 데이터과학



Model-Free Descriptive Statistics

What's Your Research Design?



Note that this might depend on the research context.

Examples of Exogenous Shock/Event for Quasi-Experiments

- Example: Impact of the access to local finance on firm formation (Gilje 2019)

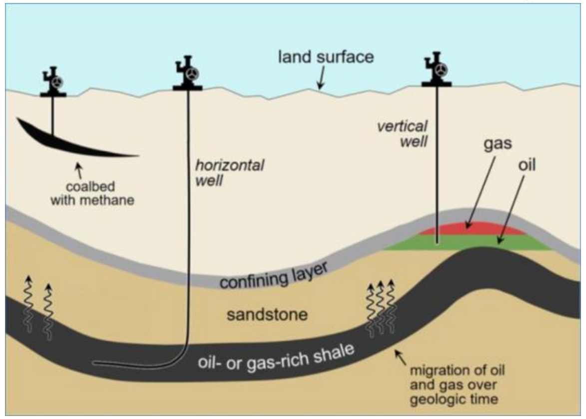
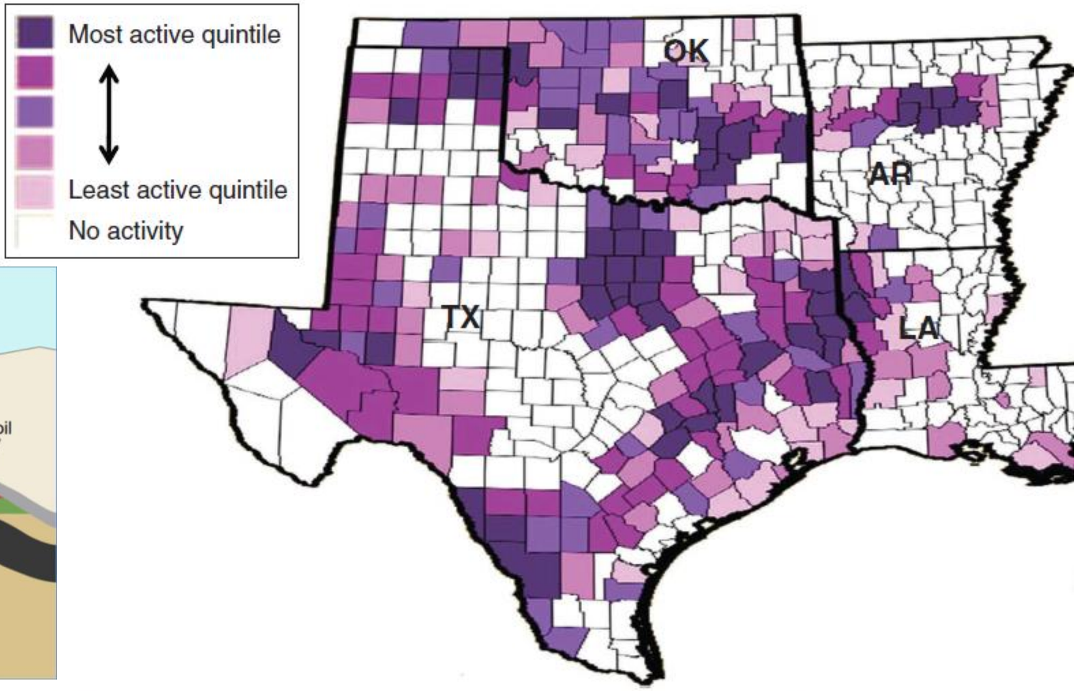


Figure 1. (Color online) Location and Intensity of Shale Activity



Notes. The figure maps the counties of the seven shale boom states included in this study: OK, TX, are counties with no shale development activity. The remaining counties are shaded based on inter of shale wells drilled through 2009.

Shale Boom **(Natural Shock)**

More deposit in local banks by oil firms

Access to Local Finance **(Treatment)**

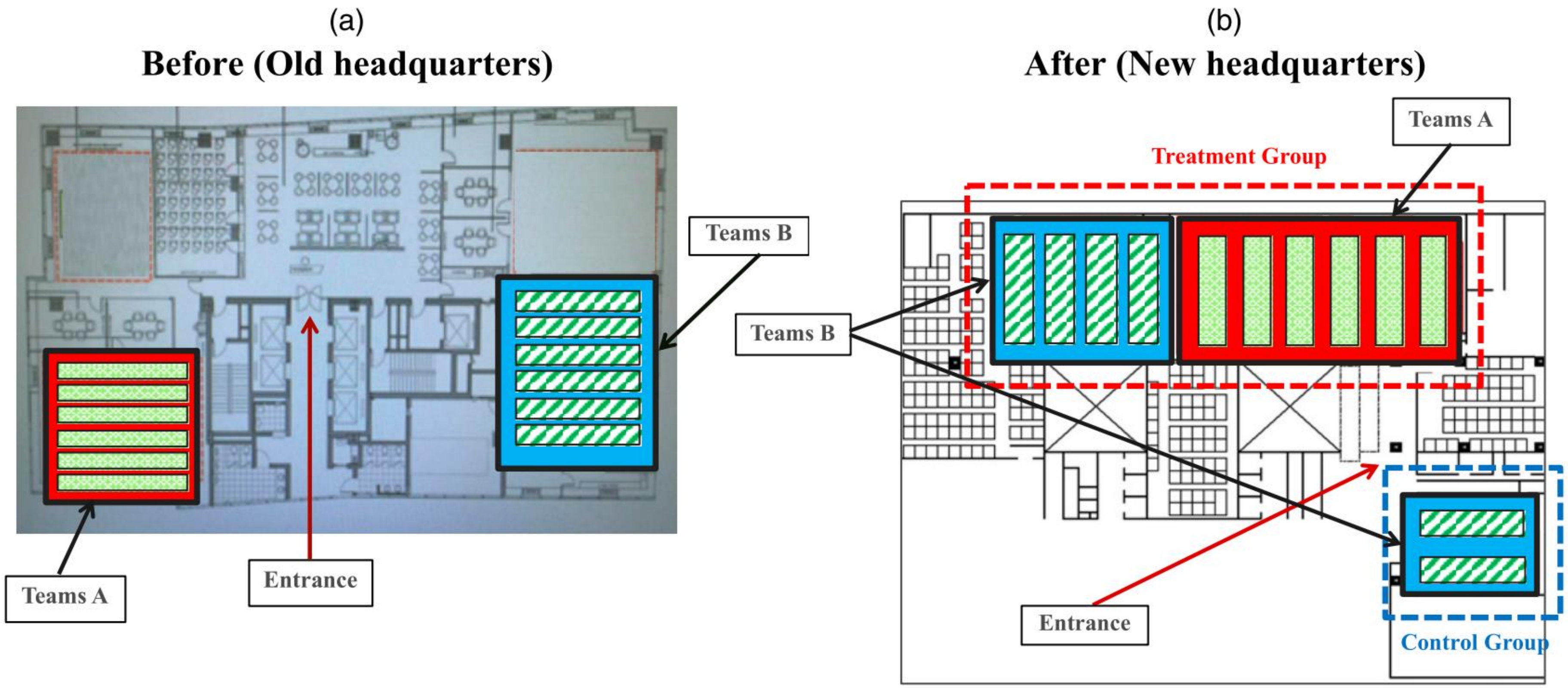
Gilje, E.P., 2019. Does Local Access to Finance Matter? Evidence from US Oil and Natural Gas Shale Booms. *Management Science*, 65(1), pp.1-18.

Examples of Exogenous Shock/Event for Quasi-Experiments

What if..? 인과추론의 데이터과학

- Example: Impact of the spatial proximity on individual-level exploration (Lee 2019)

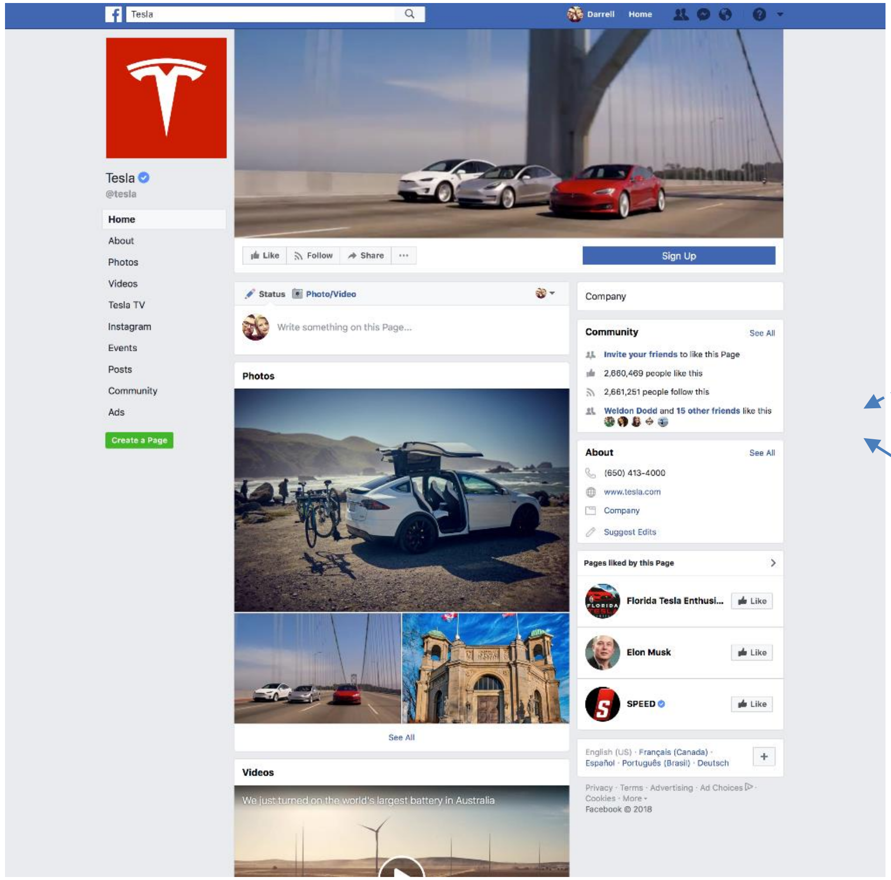
Figure 2. (Color online) Natural Experiment: Workspace Reconfiguration Due to Relocation of Headquarters



Lee, S., 2019. Learning-by-moving: can reconfiguring spatial proximity between organizational members promote individual-level exploration?. *Organization Science*, 30(3), pp.467-488.

Examples of Self-Selection for Quasi-Experiments

- Example: Effect of customers' social media participation on their visit frequency (Rishika et al. 2013)



Customers who do not participate in the firm's Facebook site

Control Group

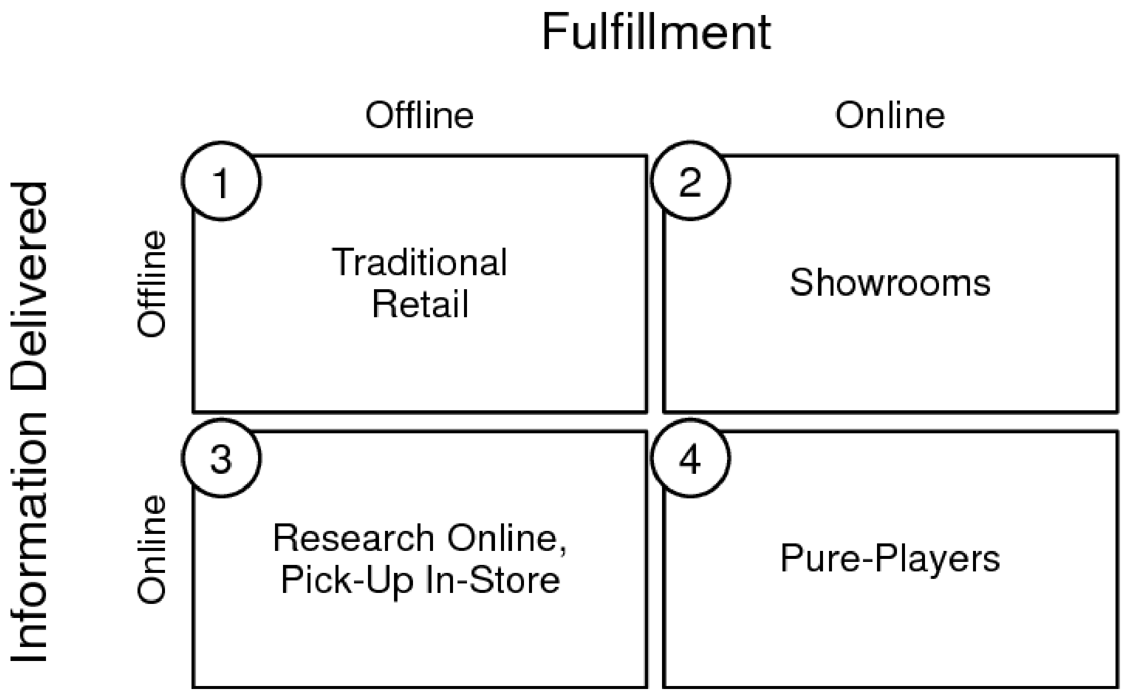
Treatment Group

Customers who actively participate in the firm's Facebook site (e.g., reply, like, etc.).

Rishika, R., Kumar, A., Janakiraman, R. and Bezawada, R., 2013. The Effect of Customers' Social Media Participation on Customer Visit Frequency and Profitability: An Empirical Investigation. *Information Systems Research*, 24(1), pp.108-127.

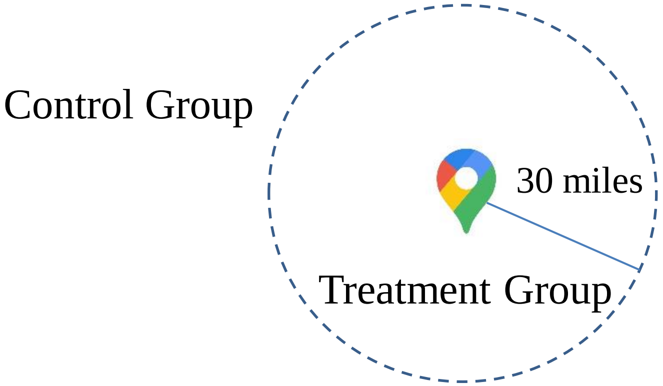
Examples of Self-Selection for Quasi-Experiments

- Example: Effect of offline showrooms on demand generation and operational efficiency (Bell et al. 2018)



Bell, D.R., Gallino, S. and Moreno, A., 2018. Offline showrooms in omnichannel retail: Demand and operational benefits. *Management Science*, 64(4), pp.1629-1651.

Warby Parker's showroom



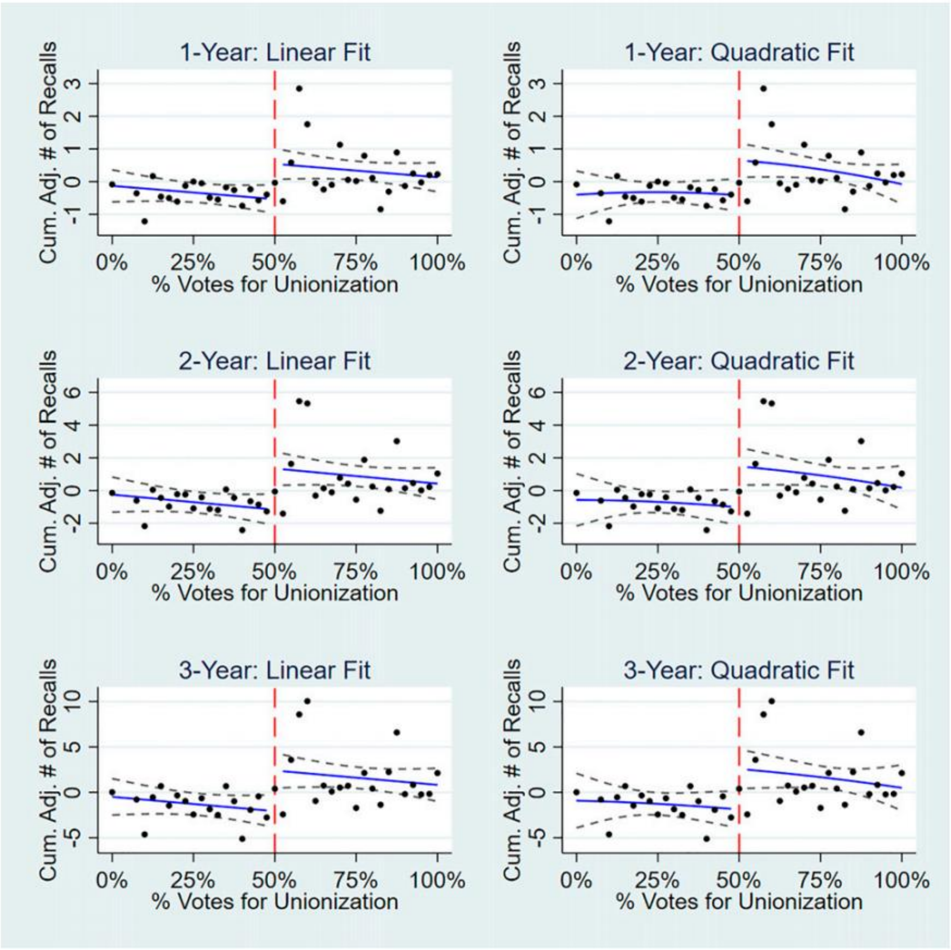
Examples of Discontinuity for Quasi-Experiments

- Example: Effect of labor unions on product quality failures (Kini et al. 2021)

“We compare the post-election frequency of product recalls for firms whose employees vote to become unionized with those whose employees vote against forming a union.

To circumvent the endogenous nature of union election results, we employ a regression discontinuity design (RDD) methodology, which compares firms with close union victories to firms with close union losses.”

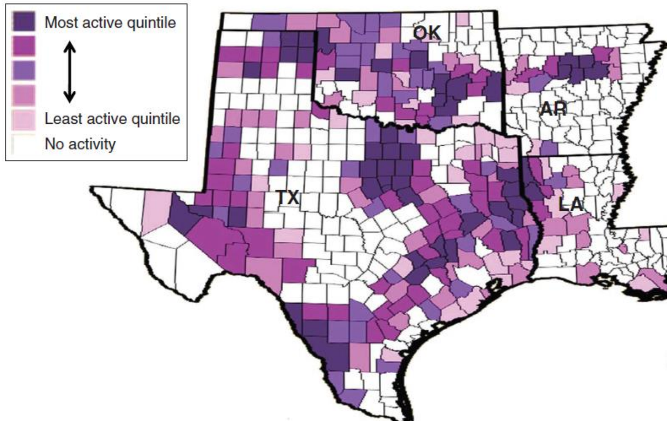
Figure 3. (Color online) Regression Discontinuity Plots

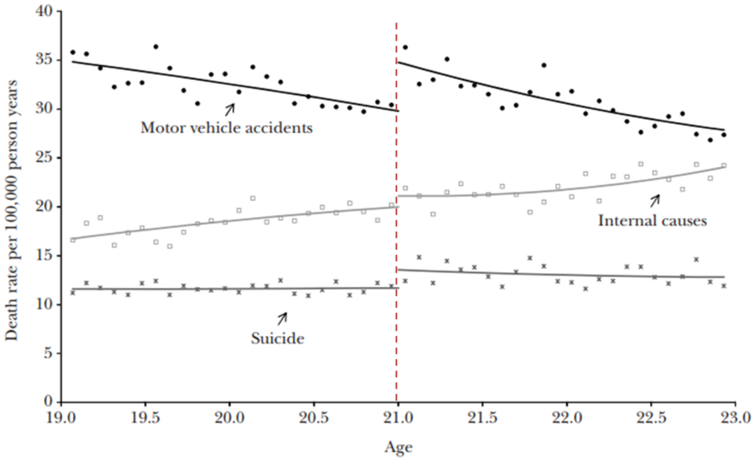


Notes. Regression discontinuity plots using fitted linear and quadratic polynomial estimates. The *x*-axis is the percentage of votes in favor of unionization. The *y*-axis is the one-, two-, and three-year period cumulative adjusted number of recalls computed as the residuals from a linear regression of cumulative recalls on industry and year fixed effects. The dots depict the average of the cumulative adjusted number of recalls in each of the 40 equally spaced bins. The fitted values and the respective 95% confidence interval are plotted based either on a global linear or quadratic polynomial function. The union elections are over 2002–2012 for which we have recalls information in the one-, two-, and three-year post-election period.

Kini, O., Shen, M., Shenoy, J. and Subramaniam, V., 2021. Labor unions and product quality failures. *Management Science*. forthcoming

Burden of Proof for *Ceteris Paribus*





Assigned randomly

Assigned by exogenous shock or arbitrary cutoff

Assigned by themselves (self-selection)

Burden of Proof and Level of Difficulty for Ceteris Paribus

Design-Based Approach to Causality

“The analyst is successful at identifying the causal effect not because of the complex statistical methods that are applied to the data, but due to the effort in developing a design before data are collected.” (Keele 2015, p. 331)

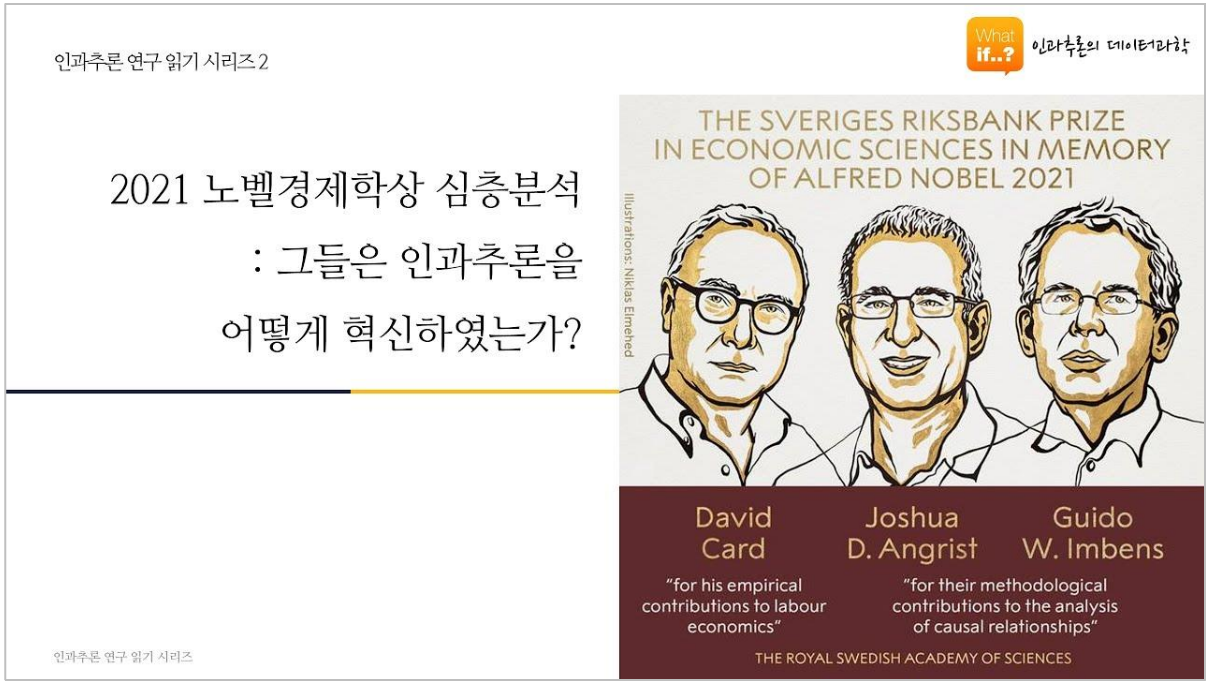
Keele, L., 2015. The Statistics of Causal Inference: A View from Political Methodology. *Political Analysis*, 23(3), pp.313-335.

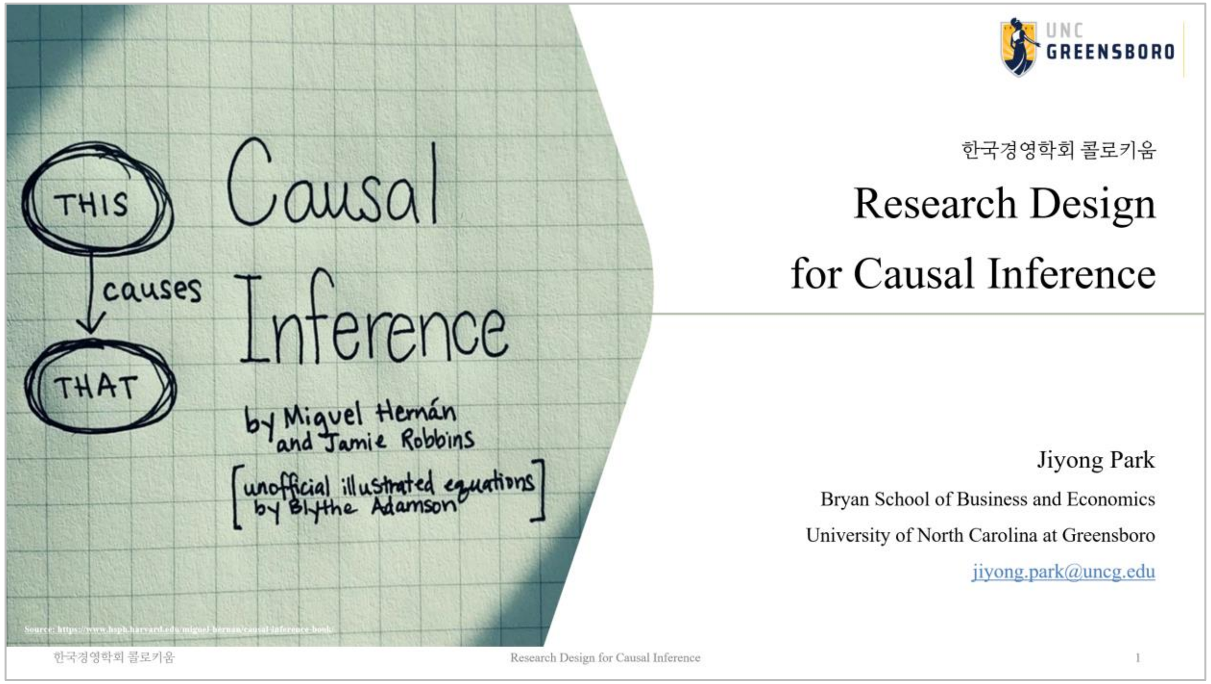
Statistical Methods for Analyzing Quasi-Experiments

- Difference-in-Differences (DID)
- Regression Discontinuity (RD)
- Propensity Score Matching (PSM) & Coarsened Exact Matching (CEM)
- Instrumental Variable (IV)
- Synthetic Control (SC)
- Etc.

Further Learning

- [2021 노벨경제학상 심층분석: 그들은 인과추론을 어떻게 혁신하였는가?]
- [[한국경영학회 특강] 인과추론을 위한 연구디자인]





Further Learning

- [Session 1-2] 잠재적결과 프레임워크 (Potential Outcomes Framework)
- [Session 1-3] 인과적 사고방식
- [Session 2-1] 인과추론을 위한 연구 디자인
- [Session 2-2] 인과추론의 정석: 무작위 통제실험 (Randomized Controlled Trial)
- [Session 2-3] 실험 아닌, 실험 같은 준실험 (Quasi-Experiment)
- [Session 2-4] 준실험 분석도구: 이중차분법 & 회귀불연속 (Difference-in-Differences & Regression Discontinuity)
- [Session 9-1] 도구 변수 (Instrumental Variable)
- [Session 9-2] 도구 변수의 활용 사례

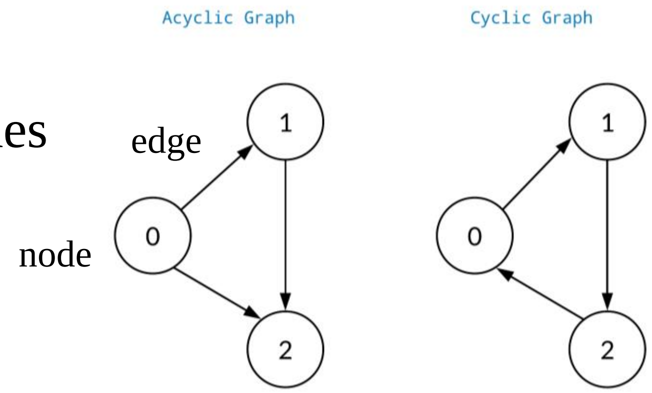
Structural Causal Model

A Graph-Based Approach

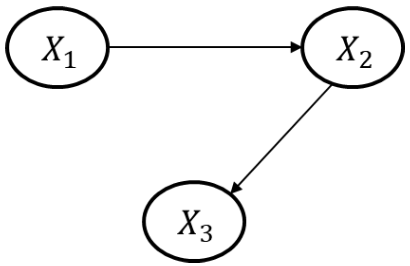
Graphical Causal Model

What if..? 인과추론의 데이터과학

- Directed Acyclic Graph (DAG)
  - *Graph:* A structure made from nodes and edges
  - *Directed*: Direction represents a causal relationship between nodes
  - *Acyclic*: No directed cycles



- Bayesian Network (Belief Network)
  - Probabilistic graphical model that represents a set of variables and their conditional dependencies



$$P(X_1, X_2, X_3) = P(X_1)P(X_2|X_1)P(X_3|X_1, X_2)$$

$$= P(X_1)P(X_2|X_1)P(X_3|X_2)$$

Graphical Causal Model

Relationship Types in Graphical Causal Model

- (Direct) Causal Effect

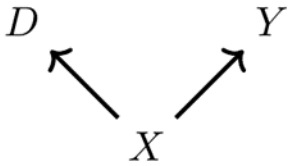
$$D \longrightarrow Y$$

- Mediator (Chain)

(Indirect Causal Effect)

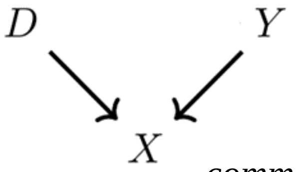
$$D \longrightarrow X \longrightarrow Y$$

- Confounder (Fork)



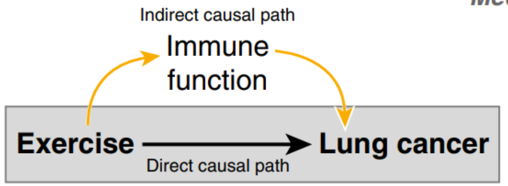
common causes

- Collider (Immorality)

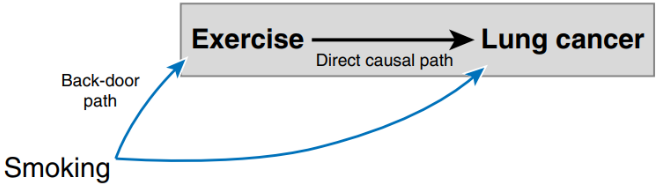


common effects

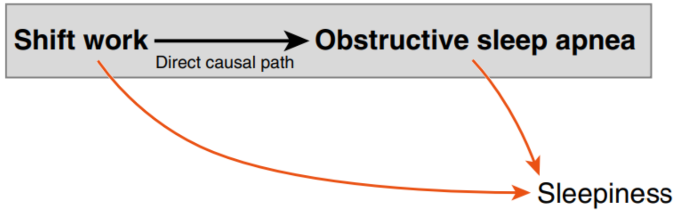
Mediation



Confounding



Collider Bias

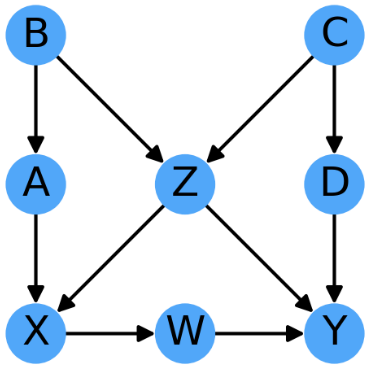


Lederer, D.J., Bell, S.C., Branson, R.D., Chalmers, J.D., Marshall, R., Maslove, D.M., Ost, D.E., Punjabi, N.M., Schatz, M., Smyth, A.R. and Stewart, P.W., 2019. Control of confounding and reporting of results in causal inference studies. Guidance for authors from editors of respiratory, sleep, and critical care journals. *Annals of the American Thoracic Society*, *16*(1), pp.22-28.

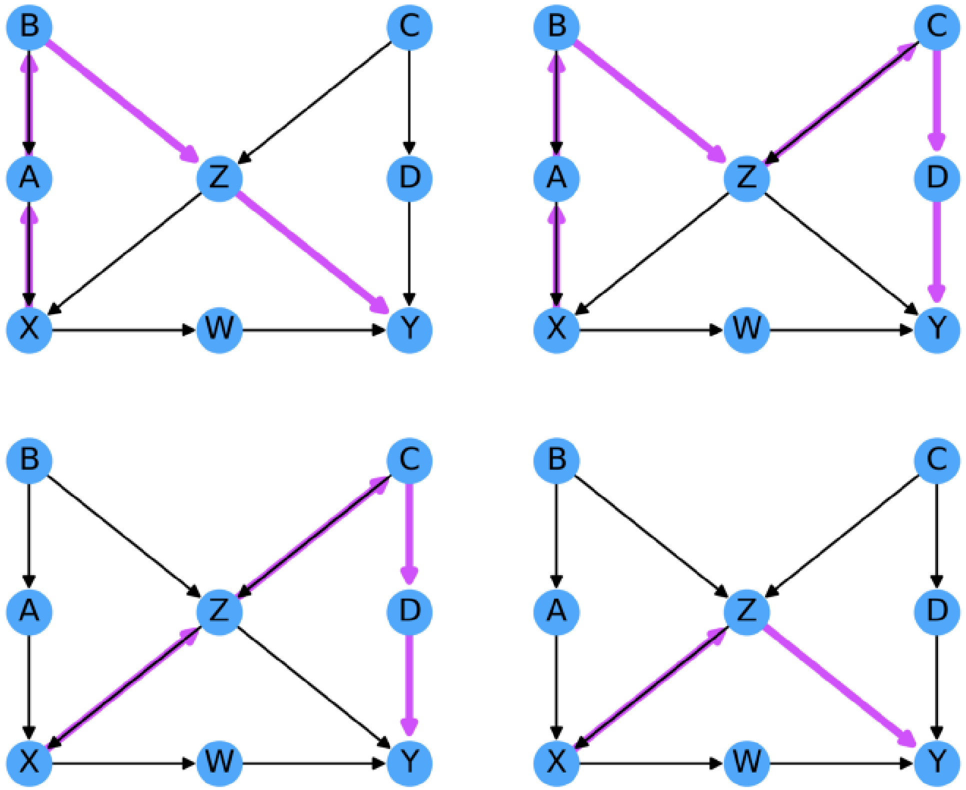
Association in Graphical Causal Model

What if..? 인과추론의 데이터과학

- Two nodes are associated if they share the same information.
  - Information of nodes flows in the direction of edges.
- The paths for noncausal associations (except direct and indirect causal effects) are called backdoor paths.



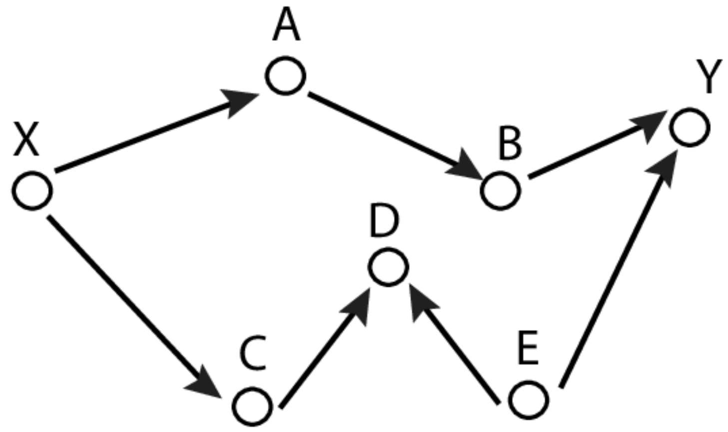
Examples of backdoor paths between X and Y



Association in Graphical Causal Model

- X and Y are d-separated if there is no information flow (no association) between X and Y.
  - All paths between X and Y are blocked.

- X and Y are d-connected if there is information flow (association) between X and Y.
  - Not all paths between X and Y are blocked.



d-separation	
Implied	Not implied
$(X \perp Y \mid A)$	$(X \perp Y \mid C)$
$(X \perp Y \mid A\ B)$	$(X \perp Y \mid C\ E)$
$(X \perp Y \mid B\ E)$	$(X \perp Y \mid A\ D)$

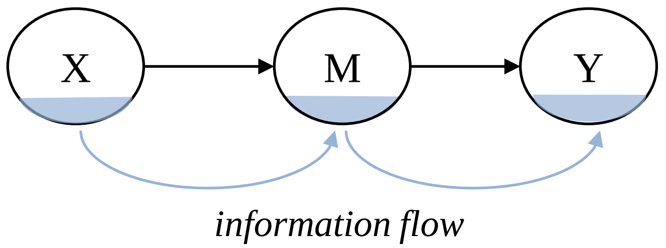
Blocking the information flow through X

= Conditioning on, or controlling for, X

Association in Graphical Causal Model

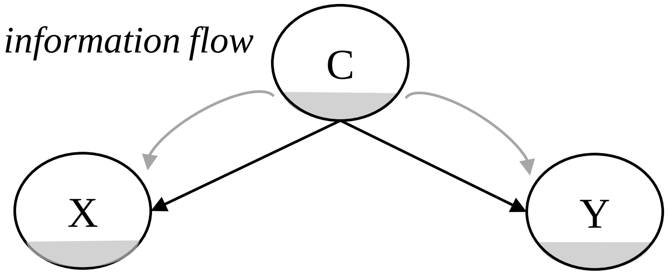
- As they are, mediators and confounders <u>do</u> establish an association between nodes, but colliders <u>do not</u>.

Mediator (Chain)



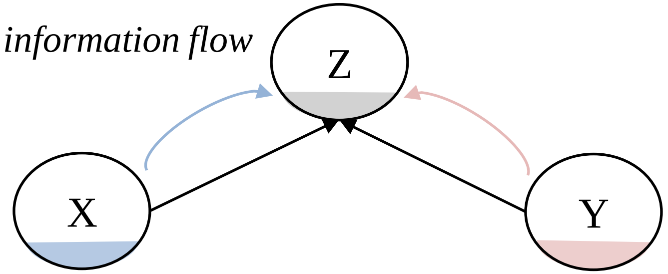
X and Y are d-connected.

Confounder (Fork)



X and Y are d-connected.

Collider (Immorality)

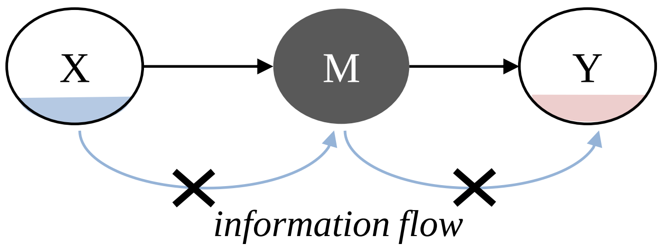


X and Y are d-separated.

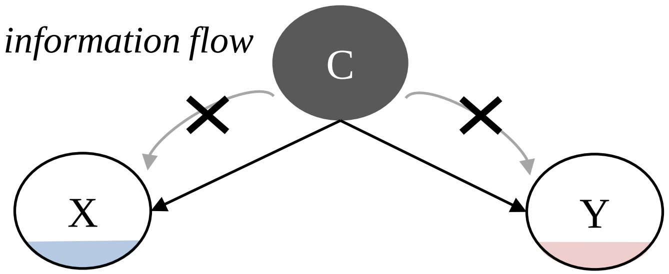
Association in Graphical Causal Model

- After conditioning on them (controlling for, or blocking), mediators and confounders <u>does not</u> establish an association between nodes, but colliders <u>do</u>.

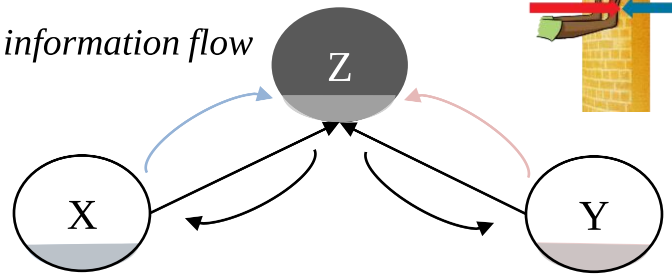
Mediator (Chain)

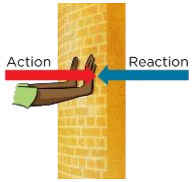


Confounder (Fork)



Collider (Immorality)





X and Y are d-separated.

To estimate the *direct* causal effect of X on Y, <u>mediators should be blocked</u>.

X and Y are d-separated.

To estimate the *direct* or *indirect* causal effect of X on Y, <u>confounders should be blocked</u>.

X and Y are d-connected.

To estimate the *direct* or *indirect* causal effect of X on Y, <u>colliders should NOT be blocked</u>.

Structural Causal Model (SCM)

Structural Causal Model $\mathscr{M} = \langle \mathbf{V}, \mathbf{U}, \mathbf{F}, P(\mathbf{u}) \rangle$

- $\mathbf{V}$: A set of endogenous (observable) variables.
- $\mathbf{U}$: A set of exogenous (latent) variables.
- $\mathbf{F}$: A set of structural equations $\{f_{V_i}\}_{V_i \in \mathbf{V}}$ determining the value of $V_i \in \mathbf{V}$, where $V_i \leftarrow f_{v_i}(PA_{V_i}, U_{V_i})$ for some $PA_{V_i} \subseteq \mathbf{V}$ and $U_{V_i} \subseteq \mathbf{U}$.
- $P(\mathbf{u})$: A probability measure for $\mathbf{U}$.

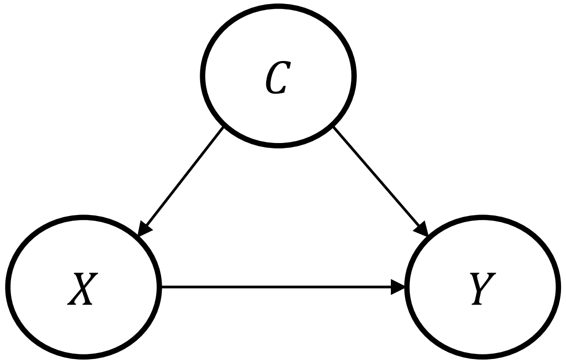
Source: Causal Inference under the Rubric of Structural Causal Model (Yonghan Jung)

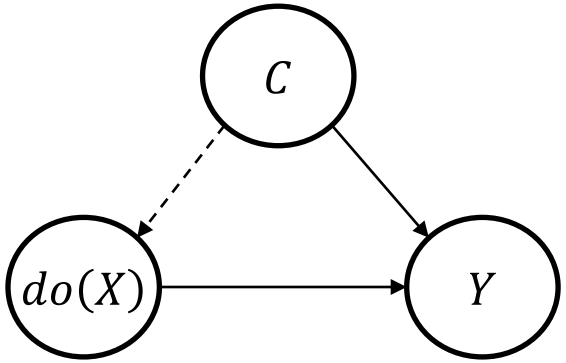
+

- **Definition of causal effect using *do-operator***
- **Identification of causal effect using *do-calculus* (with graph)**
- **Estimation of causal effect using ML models**

Definition of Causal Effect Using *do-operator*

- do(X) operator: Ignore the effects of any ancestors (parents) of X
- Causal effect of X on Y = $P\big(Y|do(X)\big)$





$P(Y|X)$ $\neq$ $P\big(Y|do(X)\big)$

Conditional Distributions | *Interventional Distributions*

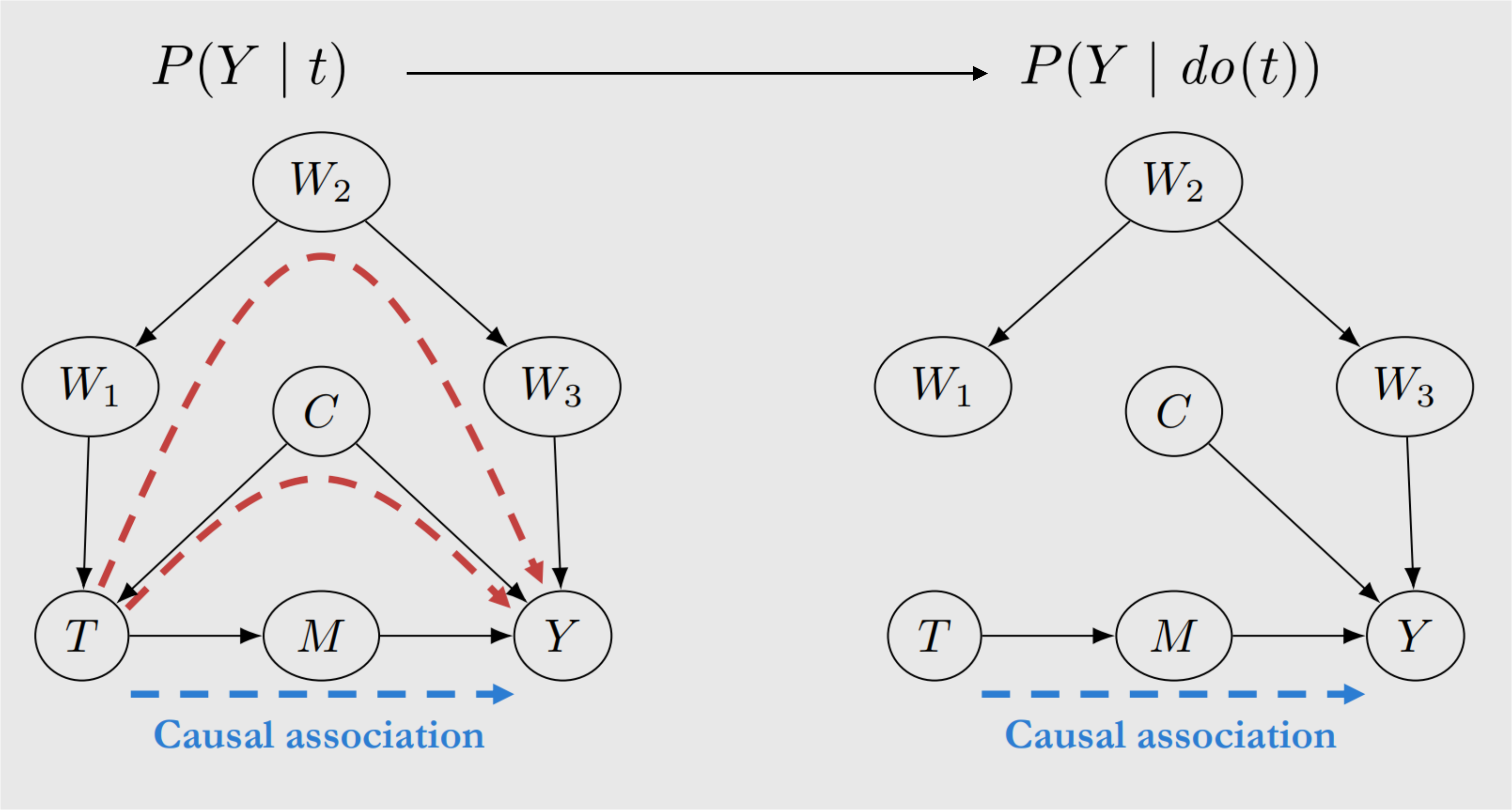
Identification

How to convert? *do-calculus*

Definition of Causal Effect Using *do-operator*

What if..? 인과추론의 데이터과학

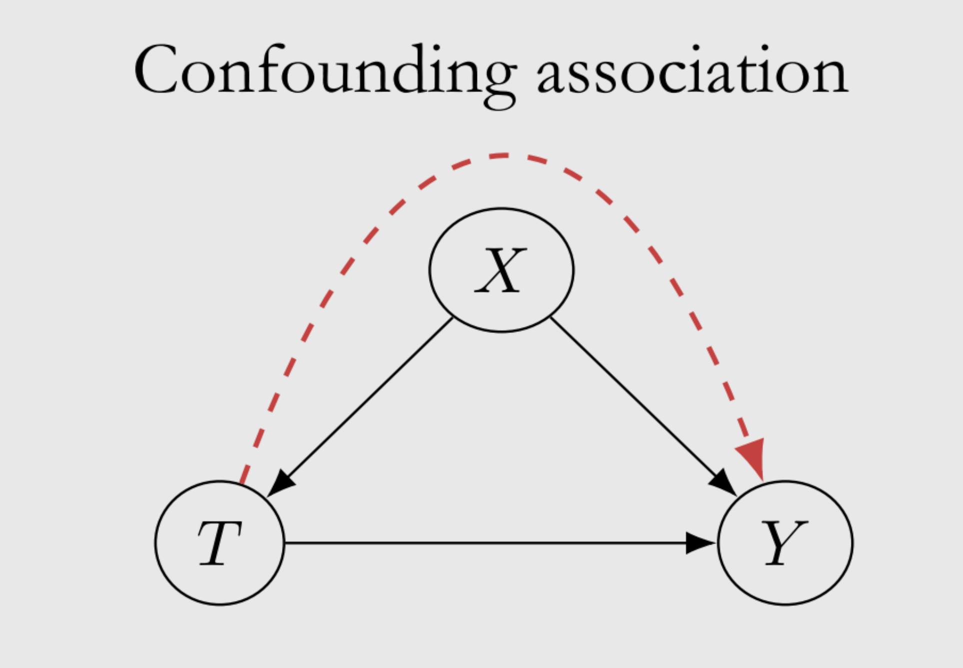
Causal Effect

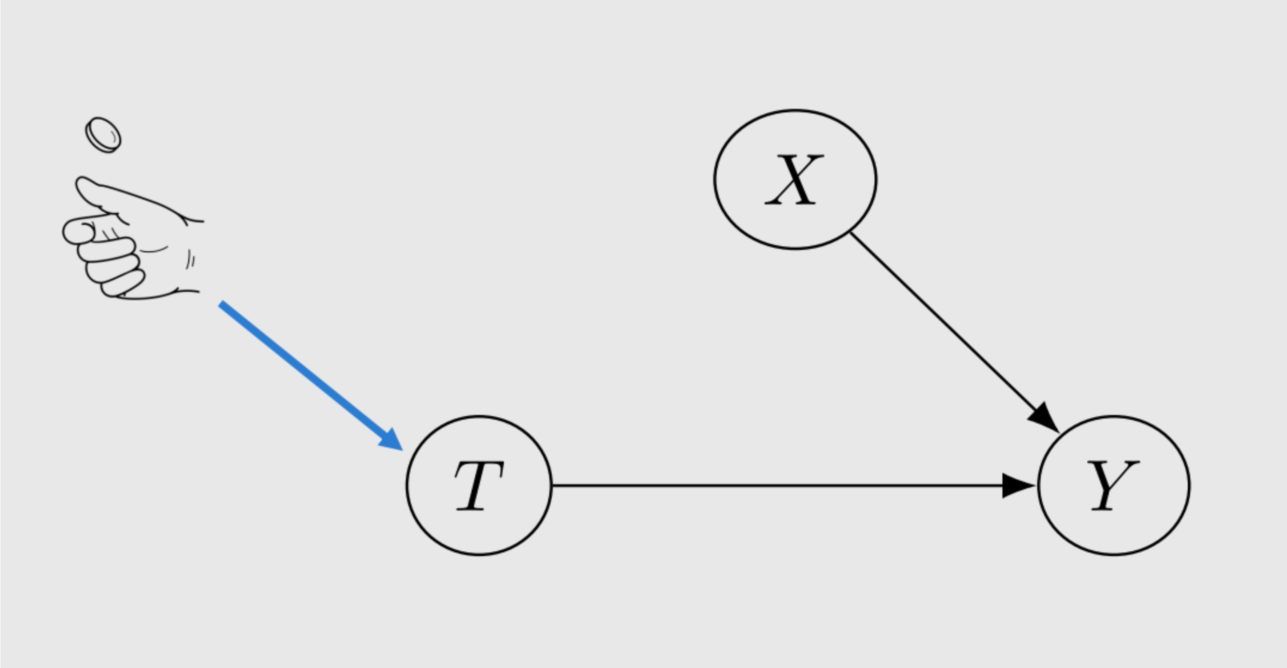


Source: Brady Neal's lecture notes
(https://www.bradyneal.com/causal-inference-course)

Revisiting Random Assignment

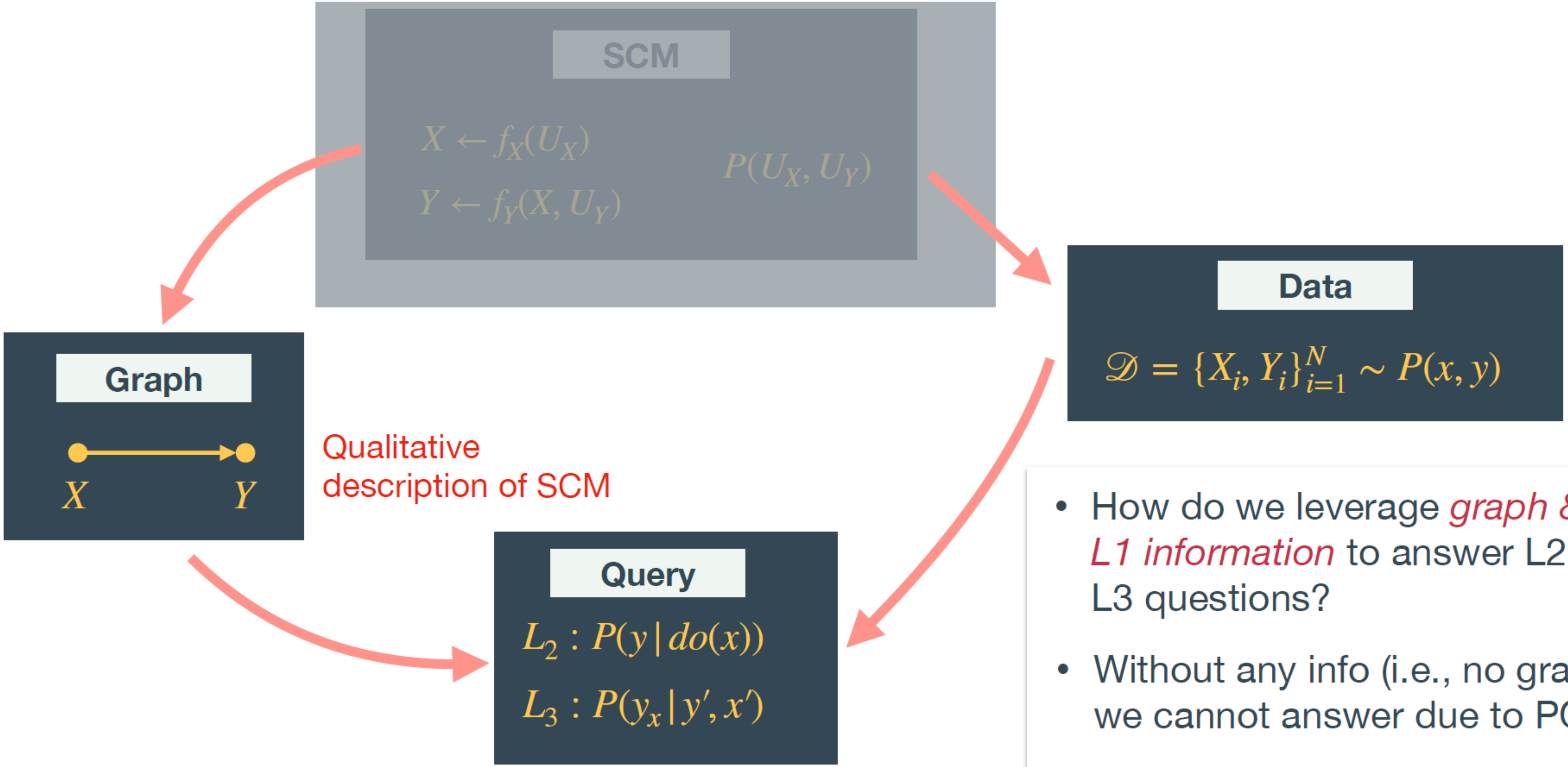
- Random assignment of the treatment T acts like the do(T) operator.





Source: Brady Neal's lecture notes
(https://www.bradyneal.com/causal-inference-course)

Causal Inference under the Rubric of SCM

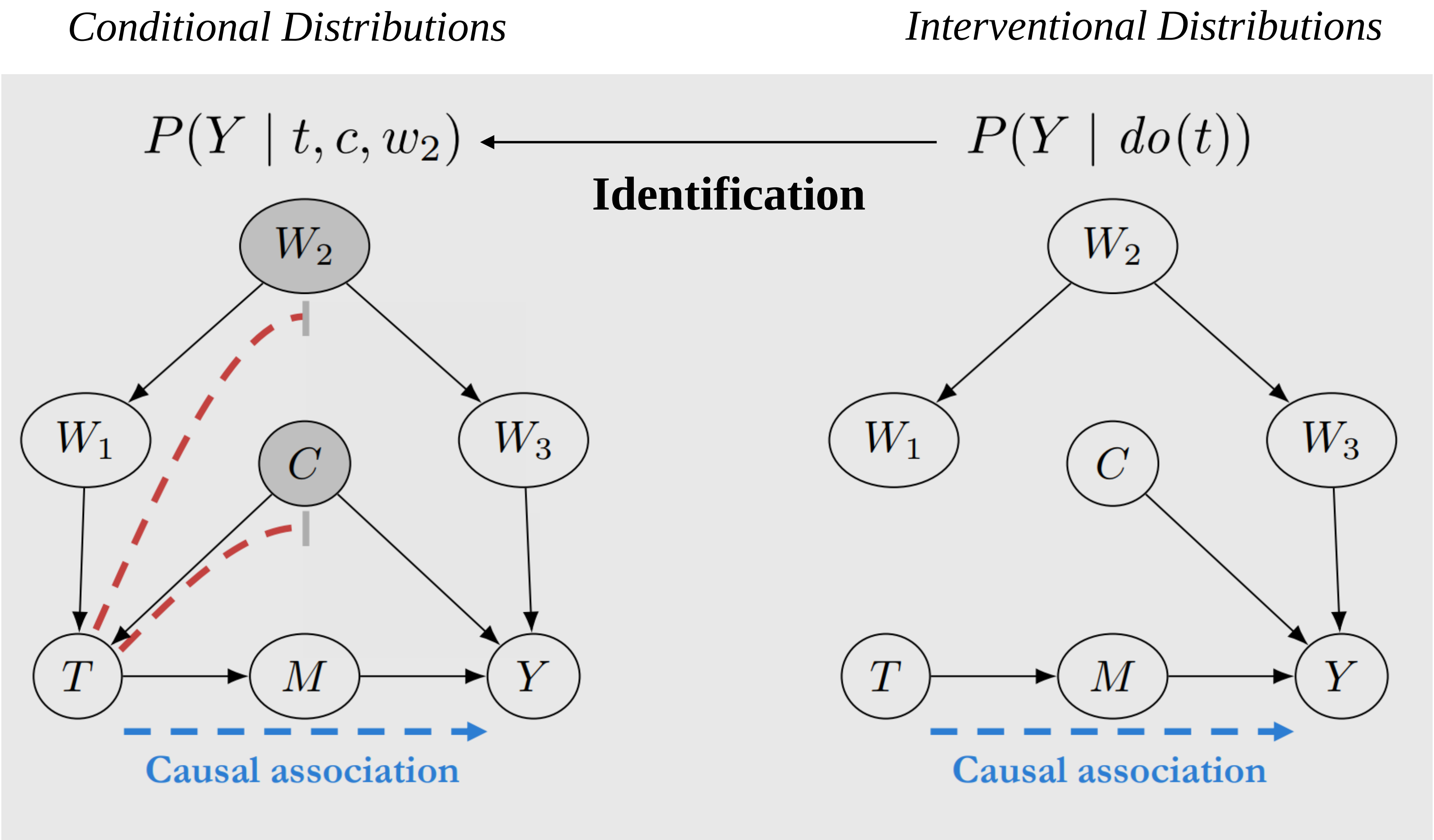


- How do we leverage *graph & L1 information* to answer L2, L3 questions?
- Without any info (i.e., no graphs), we cannot answer due to PCH)

Source: Causal Inference under the Rubric of Structural Causal Model (Yonghan Jung)

Identification of Causal Effect

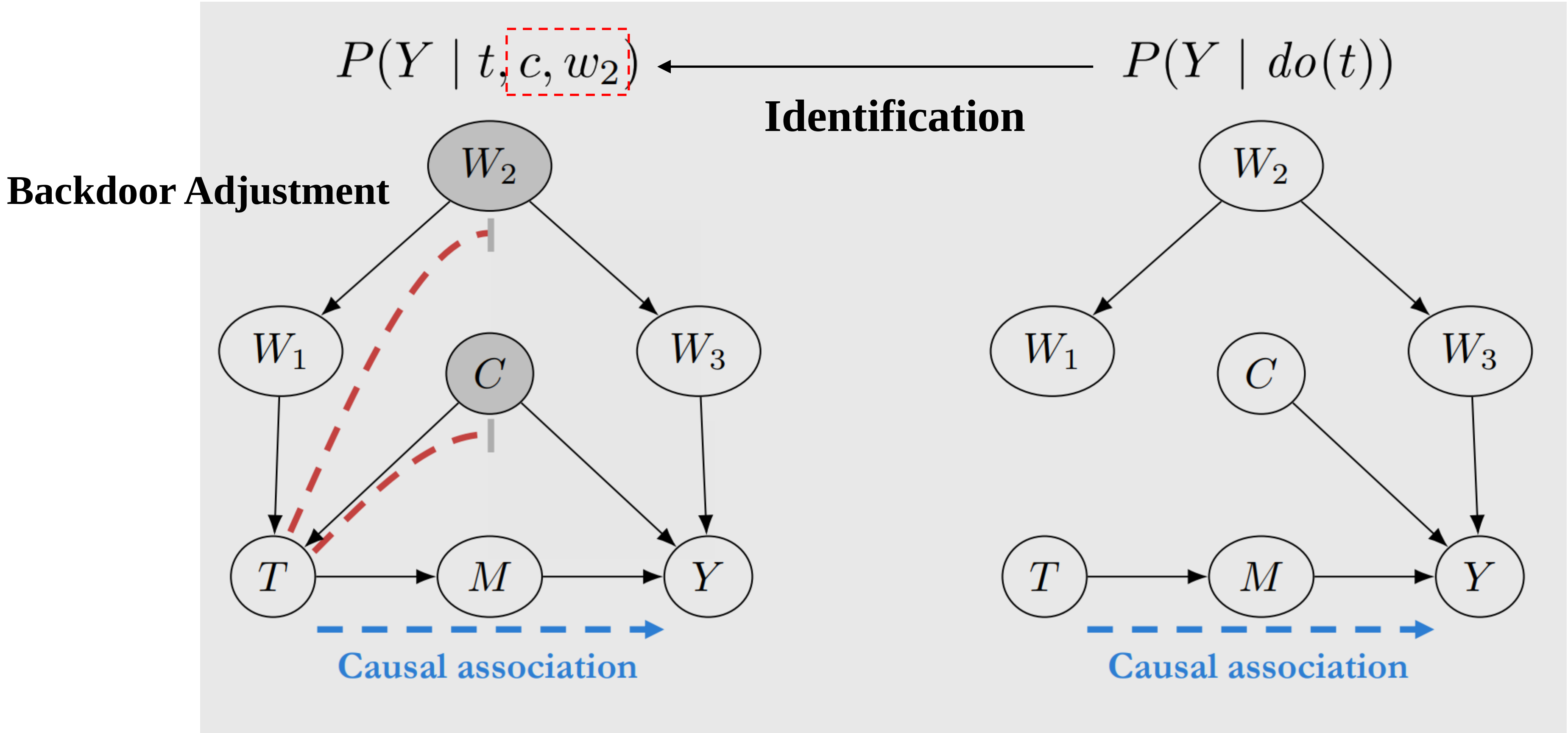
- Transforming interventional distributions (with do-operator) into estimable conditional distributions



Source: Brady Neal's lecture notes (https://www.bradyneal.com/causal-inference-course)

Identification of Causal Effect

- Backdoor criterion: A set of variables that block all paths, but the causal association, between X and Y
- Backdoor adjustment: Conditioning on variables that satisfy the backdoor criterion



Source: Brady Neal's lecture notes
(https://www.bradyneal.com/causal-inference-course)

Identification of Causal Effect Using *do-calculus* (with Graph)

What if..? 인과추론의 데이터과학

- Generalizing identification beyond the backdoor criterion

Motivated by Front-door example, Pearl [1995] developed three rules that can be used for identifying causal effect from a graph w/ $Y_x \not\perp\!\!\!\perp X \mid Z$.

Rule 1 (conditional independence):

$$(Y \perp\!\!\!\perp Z \mid X, W)_{G_{\overline{X}}} \Rightarrow P(y \mid do(x), z, w) = P(y \mid do(x), w)$$

Rule 2 (Doing/seeing interchange):

$$(Y \perp\!\!\!\perp Z \mid X, W)_{G_{\overline{X}\underline{Z}}} \Rightarrow P(y \mid do(x), do(z), w) = P(y \mid do(x), z, w)$$

Rule 3 (conditional independence for interventions)

$$(Y \perp\!\!\!\perp Z \mid X, W)_{G_{\overline{X}, \overline{Z \setminus An(W)_{G_{\overline{X}}}}}} \Rightarrow P(y \mid do(x), do(z), w) = P(y \mid do(z), w)$$

Source: Causal Inference under the Rubric of Structural Causal Model (Yonghan Jung)

Identification of Causal Effect

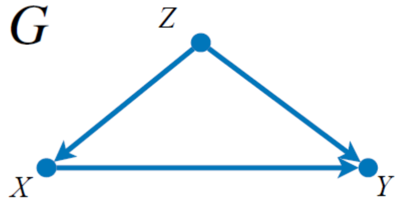
With the current scientific knowledge (encoded as a graph) about the problem (2) and the available distribution (3), can we answer the research question (1)?

① query

$$Q = P_{\mathbf{x}}(\mathbf{y}) \equiv P(\mathbf{y} \mid do(\mathbf{x}))$$

② graph

G



③ probability

$$P(\mathbf{V})$$

$\boxed{\text{ID}\,(G, P, Q)}$ → solution

yes / no

Causal Functional

$$P_{\mathbf{x}}(\mathbf{y}) = f(P)$$

Source: Causal Inference under the Rubric of Structural Causal Model (Yonghan Jung)

Estimation of Causal Effect

ID $\left(G, P, P_{\mathbf{x}}(\mathbf{y})\right)$

With (1) a causal functional $f(P)$ such that $P_{\mathbf{x}}(\mathbf{y}) = f(P)$ and (2) a dataset D, can we have a reliable estimate T_N for $P_{\mathbf{x}}(\mathbf{y})$?

1 Causal Functional

$$P(y \mid do(x)) = f(P)$$

2 Data

$$D = \{\mathbf{V}_{(i)}\}_{i=1}^{N} \sim P$$

Estimation Engine

EST$(f(P), D)$

Estimand

$$g(P) = f(P)$$

Estimator

$$T_N = \widehat{g(P)}$$

Source: Causal Inference under the Rubric of Structural Causal Model (Yonghan Jung)

Estimation Methods for Causal Effect

- Regression
- Inverse Probability Weighting (IPW)
- Doubly Robust Estimation
- Double/Debiased Machine Learning
- Meta Learner
- Covariate Shift
- Etc.

Further Learning

- [Session 7-1] 인과 그래프 (Causal Diagram)
- [Session 7-2] 인과 그래프에서의 변수 통제방법
- [Session 7-3] 인과 그래프에서의 인과추론 전략
- [Session 7/8 - 보충 1] 베이지안 네트워크 (Bayesian Network)
- [Session 7/8 - 보충 2] 베이지안 네트워크에서의 상관관계 증명
- [Session 14-2] 머신러닝을 활용한 이질적 인과관계 분석 (Heterogeneous Treatment Effect)
- [Session 16-1] 구조적 인과모형 (Structural Causal Model)
- [Session 16-2] 구조적 인과모형에서의 인과추론 방법론

Bridging the Two Worlds of Causal Inference

Strengths and Limitations… and New Ways

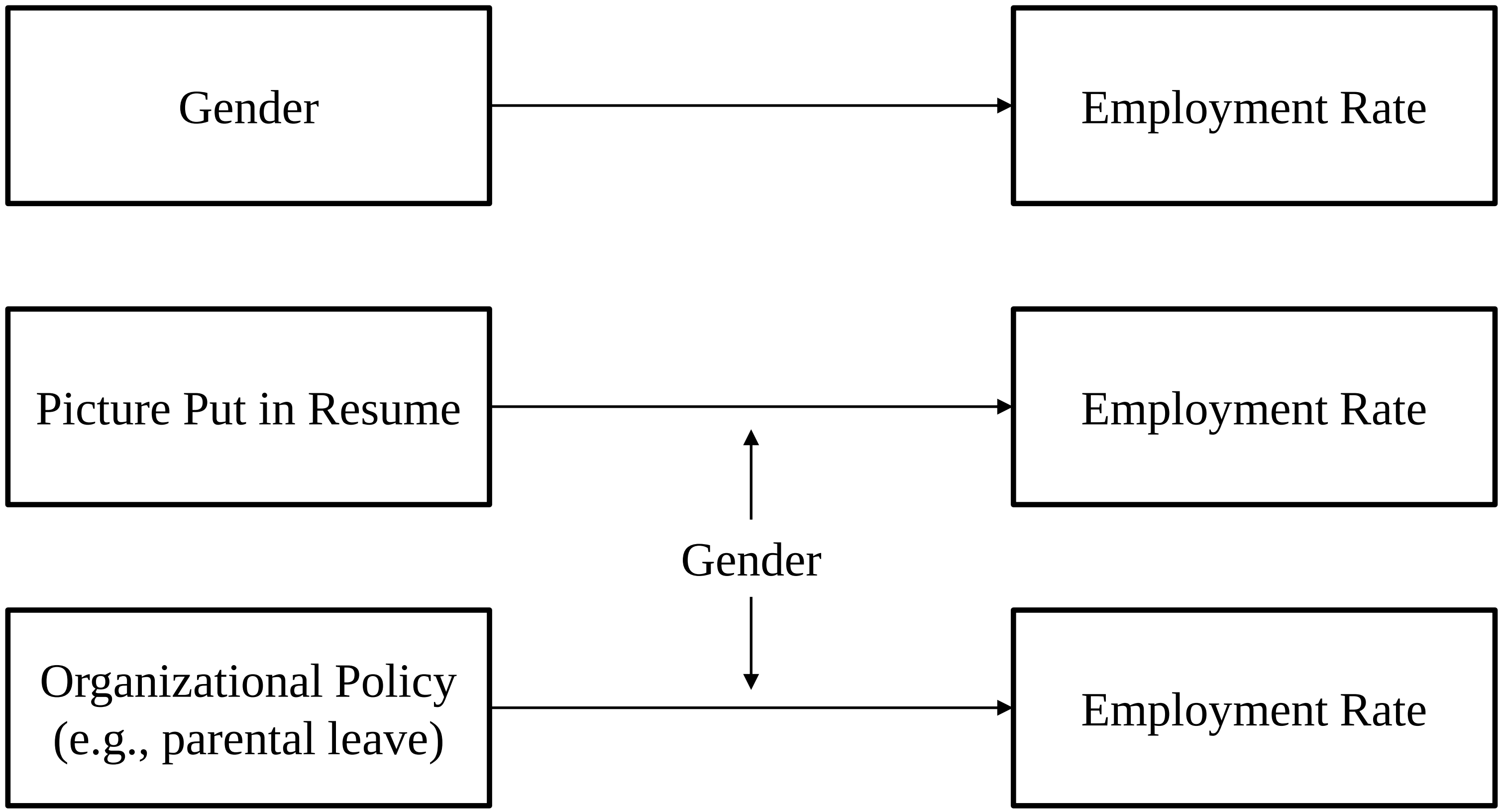
Potential Outcome Framework vs. Structural Causal Model

What if..? 인과추론의 데이터과학

	Potential Outcome Framework	Structural Causal Model
Gold Standard	Random Assignment	
Causal Inference Using Observational Data		
(1) Identification (Is it possible to estimate a causal effect?)	Research Design (Quasi-Experiment / Natural Experiment)	Backdoor Criterion / do-Calculus (based on Bayesian Network/DAG)
(2) Estimation (How to estimate a causal effect using data?)	Statistical Methods (DID, RD, Matching, IV, SC, etc.)	Statistical/Computational Methods (IPW, Doubly Robust Estimators, Double ML, etc.)

Differences (1) Manipulability

- Does gender *cause* the increase or decrease in employment?



Research Design Assumes Manipulability

- Design-based approach presumes hypothetical randomized experiments.

  “*Over 65 years ago, Haavelmo submitted the following complaint to the readers of Econometrica (1944, p. 14): “A design of experiments (a prescription of what the physicists call a ‘crucial experiment’) is an essential appendix to any quantitative theory. And we usually have some such experiment in mind when we construct the theories, although—unfortunately—most economists do not describe their design of experiments explicitly.*’”” (Angrist and Pischke 2010, p. 16)

- Causation can be defined and quantified only for the treatments that can be designed and manipulated.

  “*Without treatment definitions that specify actions to be performed on experimental units, we cannot unambiguously define causal effects of treatments*.” (Rubin 1978, p. 39)

  “*No Causation without Manipulation*” (Holland 1986, p. 959)

Angrist, J.D. and Pischke, J.S., 2010. The credibility revolution in empirical economics: How better research design is taking the con out of econometrics. *Journal of Economic Perspectives*, 24(2), pp.3-30.
Rubin, D.B., 1978. Bayesian inference for causal effects: The role of randomization. *The Annals of Statistics*, pp.34-58.
Holland, P.W., 1986. Statistics and causal inference. *Journal of the American Statistical Association*, 81(396), pp.945-960.

Manipulability Plays No Role in Causal Graph

Judea Pearl*

Does Obesity Shorten Life? Or is it the Soda?
On Non-manipulable Causes

"*Manipulability theories of causation, according to which causes earn their meaning and usefulness by transmitting change from actions to effects have had considerable intuitive appeal among scientists and philosophers [1–4]. The rise of Fisher's RCT to the "gold standard" of experimental science further entrenched manipulability as a prerequisite for causation. In some communities, this entrenchment has turned into a dogma, cast for example in the mantra "no causation without manipulation [5] that has led to cultural prohibition on labeling sex or race as "causes."*

Other research camps have been more tolerant to causal labels. In the structural causal model (SCM) framework, for example, manipulations are merely convenient means of interrogating nature, and causal relations enjoy independent existence, oblivious to external interventions [6–9]. In this framework, variables earn causal character through their capacity to sense and respond to changes in other variables. For example the variable "sex" earns the label "cause" by virtue of having responders such as "hormone content" or "height" which are gender dependent." (Pearl 2018, p. 1)

Pearl, J., 2018. Does obesity shorten life? Or is it the soda? On non-manipulable causes. *Journal of Causal Inference*, *6*(2).

Manipulability Plays No Role in Causal Graph

- Example: Causal discovery of Alzheimer's pathophysiology

Does it make sense that education causes sex (gender)?

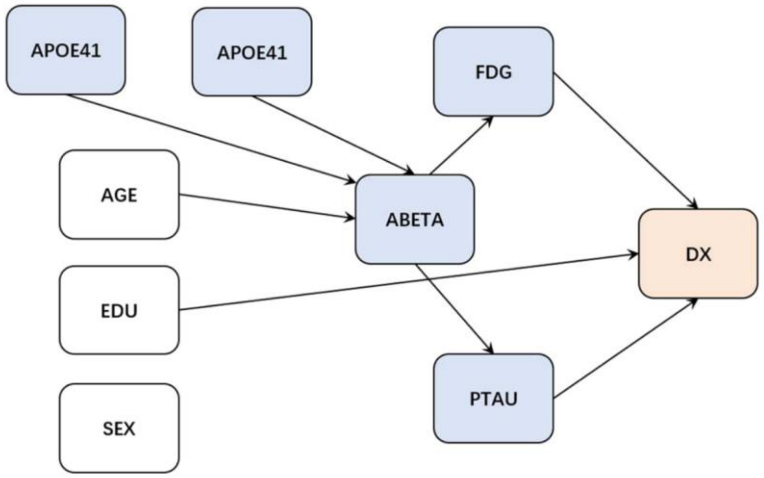
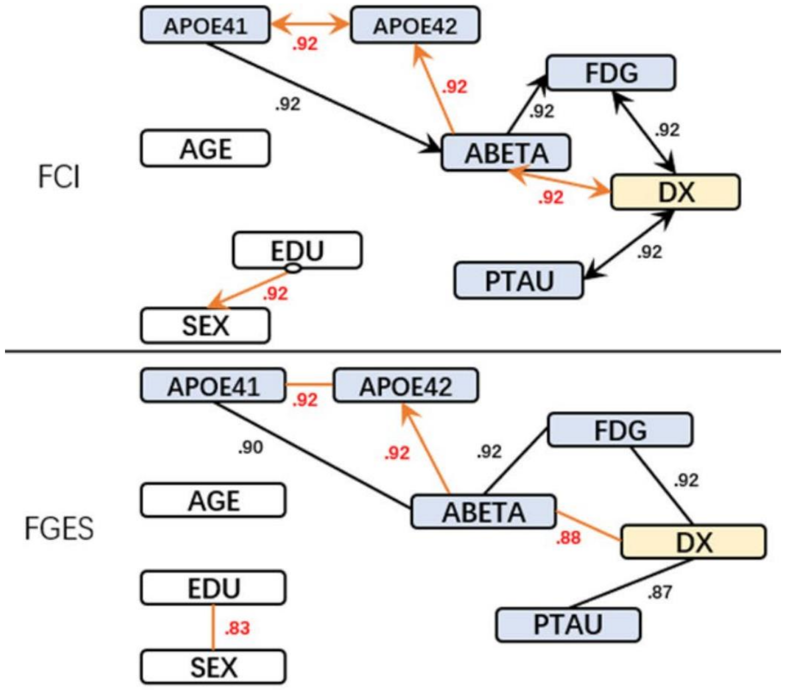
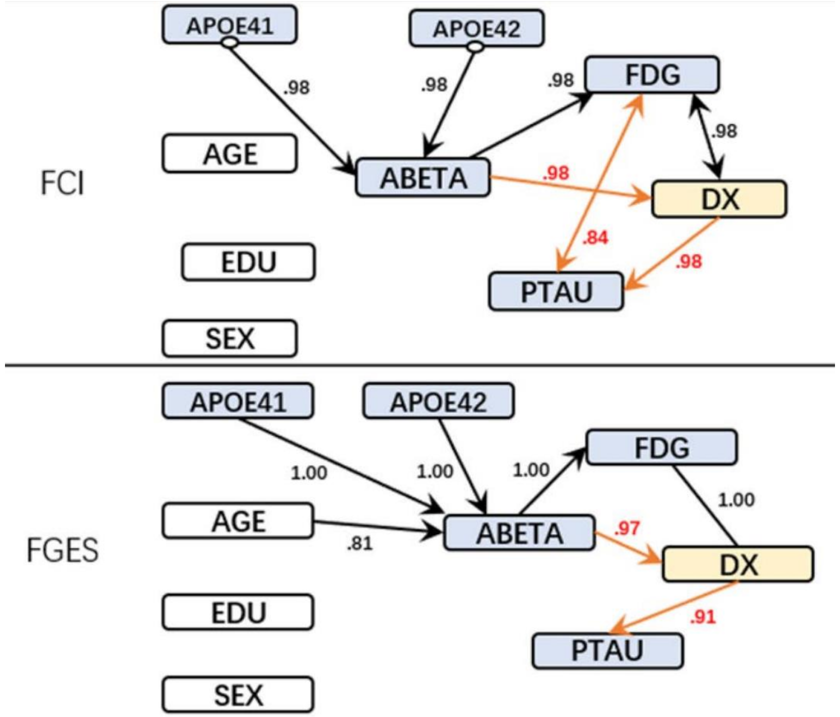


Figure 2. The "gold standard" graph.





Shen, X., Ma, S., Vemuri, P. and Simon, G., 2020. Challenges and opportunities with causal discovery algorithms: Application to Alzheimer's pathophysiology. *Scientific Reports*, *10*(1), pp.1-12.

Without accounting for manipulability

With accounting for manipulability
(Assuming that demographic variables cannot be caused by others)

Differences (2) Causal Structure/Knowledge

What if..? 인과추론의 데이터과학

- Example: Maternal status and birth outcome

  (1) Graph-based approach leveraging the causal structure/knowledge (Vahratian et al. 2005)

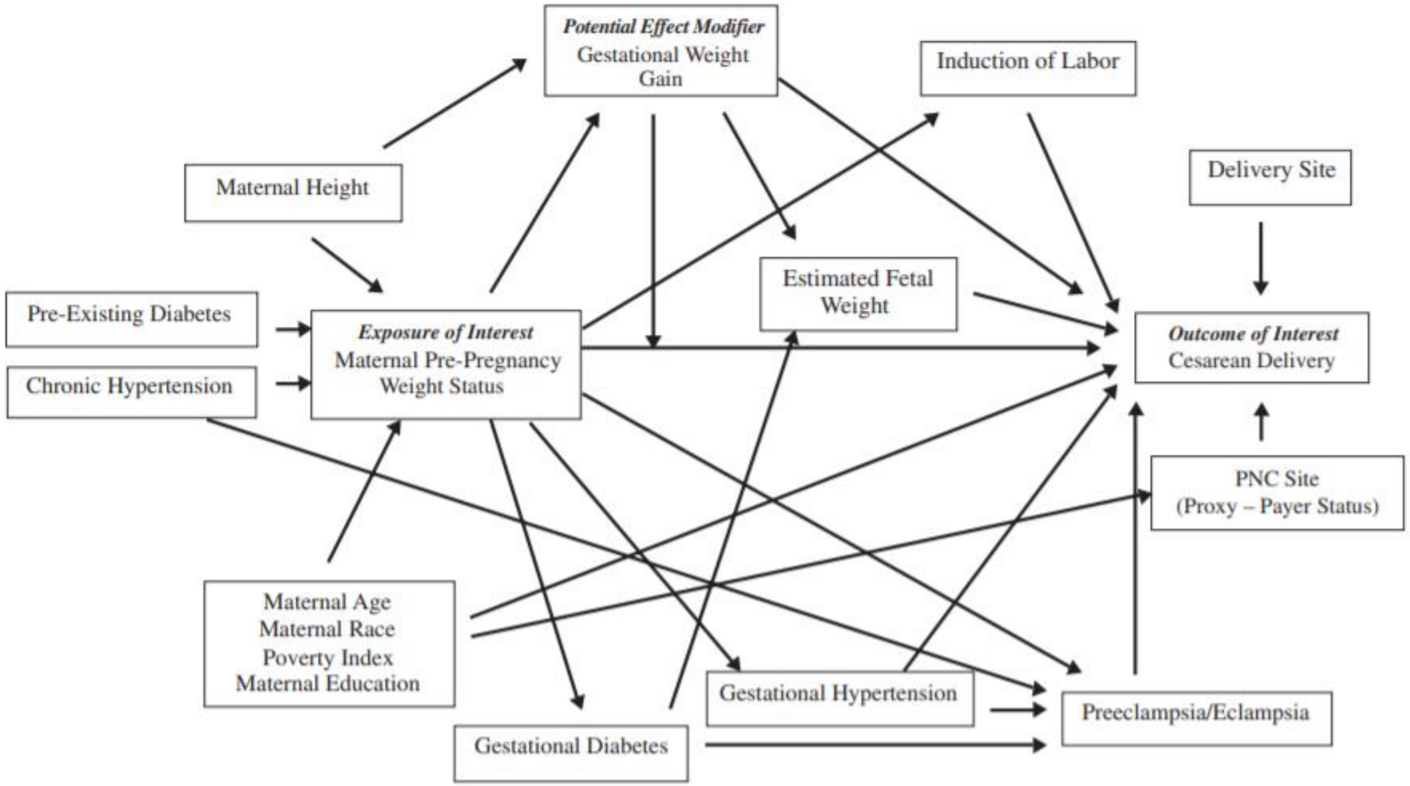


FIGURE 1. Directed acyclic graph (DAG) of the association between maternal pre-pregnancy weight status and the risk of cesarean delivery. Briefly, DAGs provide a visual method for identifying potential effect modifiers and confounders to be measured and controlled

Vahratian, A., Siega-Riz, A.M., Savitz, D.A. and Zhang, J., 2005. Maternal pre-pregnancy overweight and obesity and the risk of cesarean delivery in nulliparous women. *Annals of Epidemiology*, *15*(7), pp.467-474.

Differences (2) Causal Structure/Knowledge

인과추론의 데이터과학

- Example: Maternal status and birth outcome

(2) Design-based approach without imposing the causal structure/knowledge (Torche 2011)

"*This article examines the effect of one such condition - prenatal maternal stress - on birth weight, an early outcome shown to affect cognitive, educational, and socioeconomic attainment later in life.* ***Exploiting a major earthquake as a source of acute stress and using a difference-in-difference methodology****, I find that maternal exposure to stress results in a significant decline in birth weight and an increase in the proportion of low birth weight.*" (Torche 2011, p. 1473)

Torche, F., 2011. The effect of maternal stress on birth outcomes: Exploiting a natural experiment. *Demography*, *48*(4), pp.1473-1491.

Policy-Based vs. Knowledge-Based Causation

Manipulability → **Causal Structure/Knowledge**

My Own View

Source: Kosuke Imai's Lecture Note (Harvard U)

- Potential outcomes are useful when thinking about treatment assignment mechanism ⇝ experiments, quasi-experiments
- DAGs are useful when thinking about the entire causal structure ⇝ complex causal relationships
- Both are better suited for causal inference than the standard regression framework

https://imai.fas.harvard.edu/teaching/files/cause.pdf

Differences (3) Programmability

Manipulability

Causal Structure/Knowledge

Programmability

Automation

Algorithm

Causal Structure/Path are Required to be Programmed

- Identification under the Structural Causal Model has been mostly automated (i.e., algorithmized).

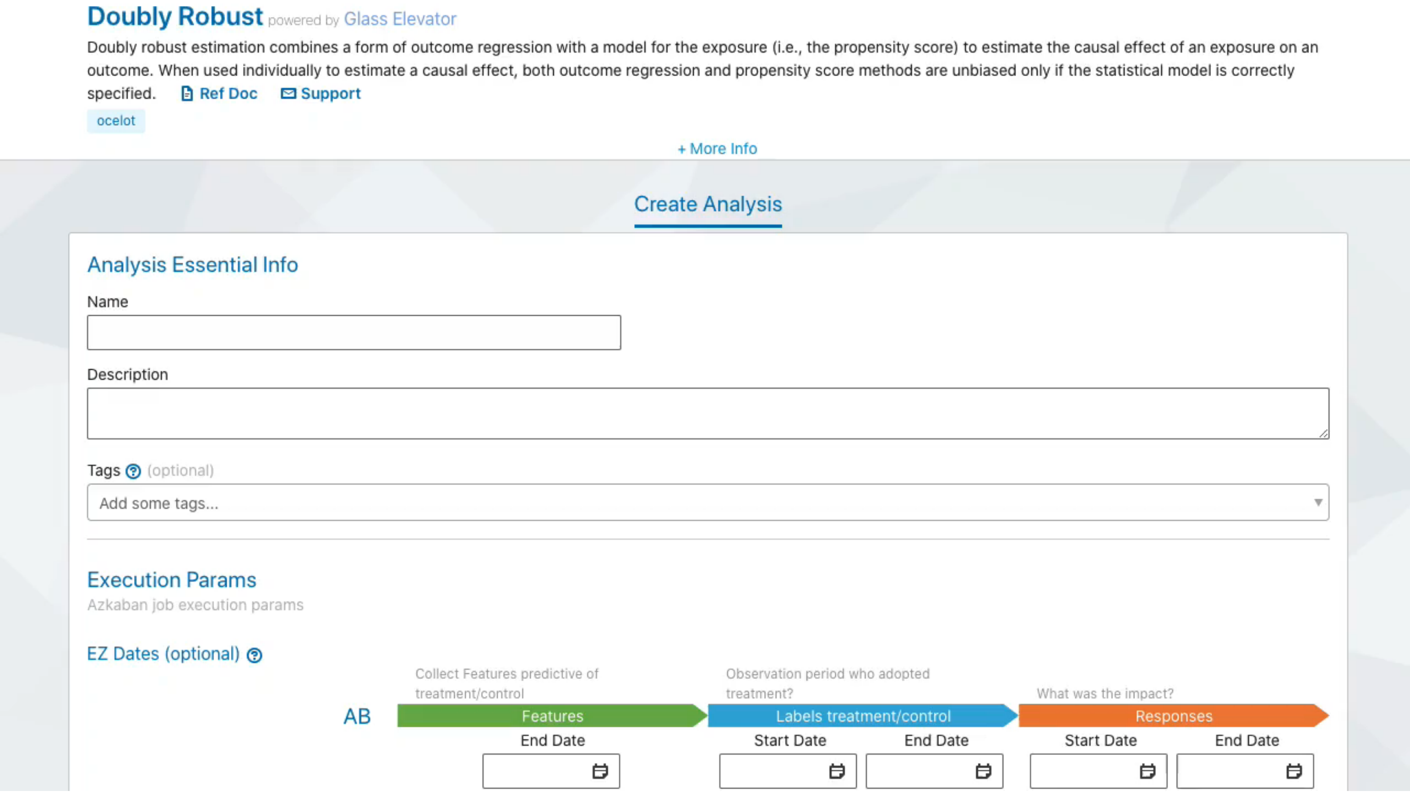
https://www.youtube.com/watch?v=921o8h5t32k
http://www.dagitty.net/dags.html

Causal Structure/Path are Required to be Programmed

- Estimation methods under the Structural Causal Model have begun to be automated (i.e., algorithmized).
  - Example: Doubly robust estimation at LinkedIn (Bojinov et al. 2020)

"At LinkedIn, we dramatically decreased analysis time while increasing quality by automating portions of the work…

To further improve reproducibility, productivity, and trust, we built a causal inference web platform, hosting all four categories of methods described in the case studies. Data scientists can execute an analysis at a click, with backend automation of computation and validation."



Bojinov, I., Chen, A. and Liu, M., 2020. The importance of being causal. Harvard Data Science Review 2.3.
https://hdsr.mitpress.mit.edu/pub/wjhth9tr/release/1

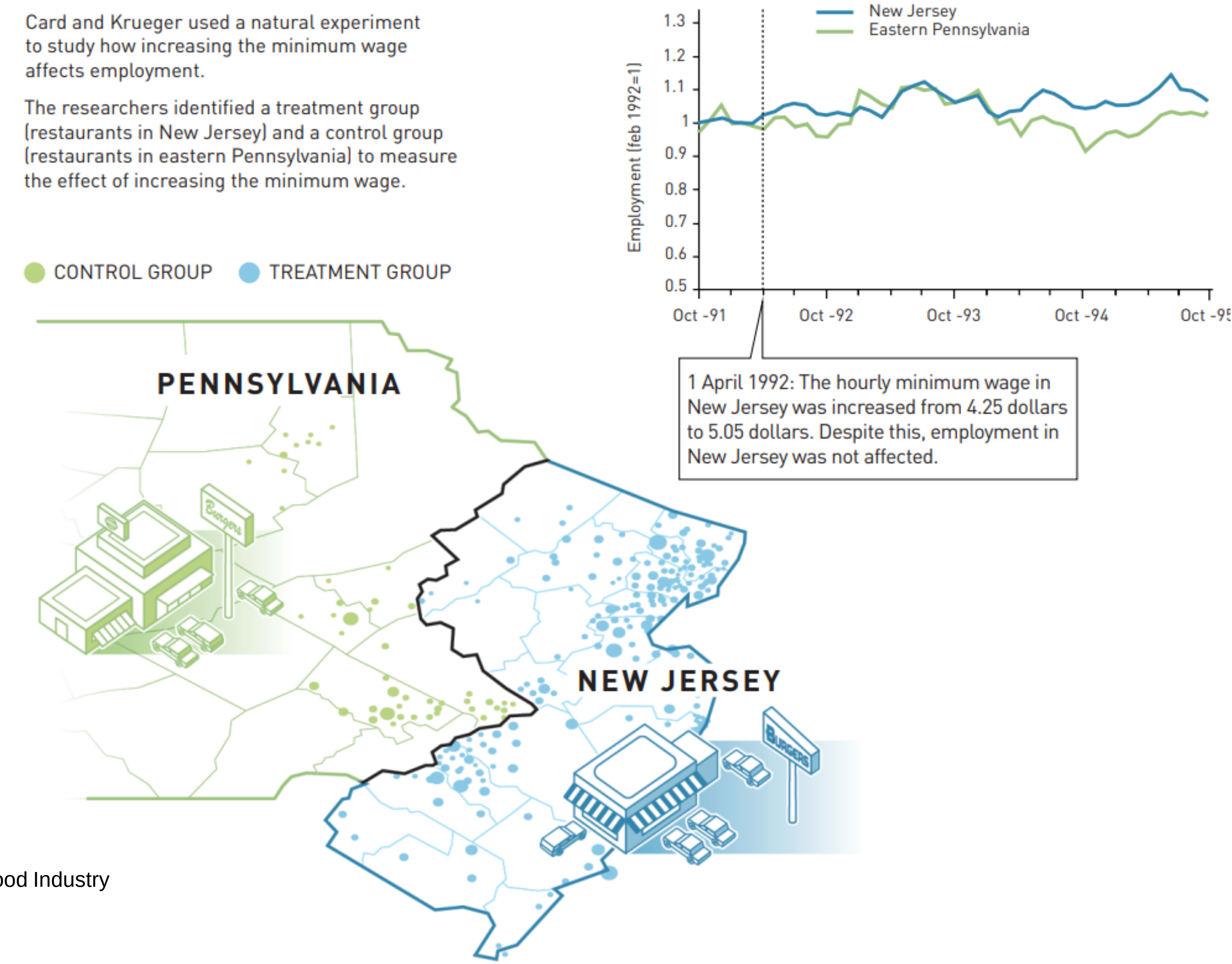
https://assets.pubpub.org/0nvixam8/01595007182461.mp4

Limitation of Potential Outcome: Not Programmable

- Randomized control trial can be programmable, which is why A/B tests are dominating IT industries for causal inference.

- However, quasi-experiments are hardly programmable, as research designs depend on the context (not typical), and researcher’s involvement is inevitable.

Can we apply the same research design on the minimum wage increase in the Pennsylvania and New Jersey to South Korea?



Card, D. and Krueger, A.B., 1994. Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania. *American Economic Review*, *84*(4), pp.772-93.

Programmable Potential Outcome: Synthetic Control

- Synthetic control ≅ Predicted counterfactual
  - Example: Impact of Reunification on West Germany (see Abadie 2021 for a comprehensive review)

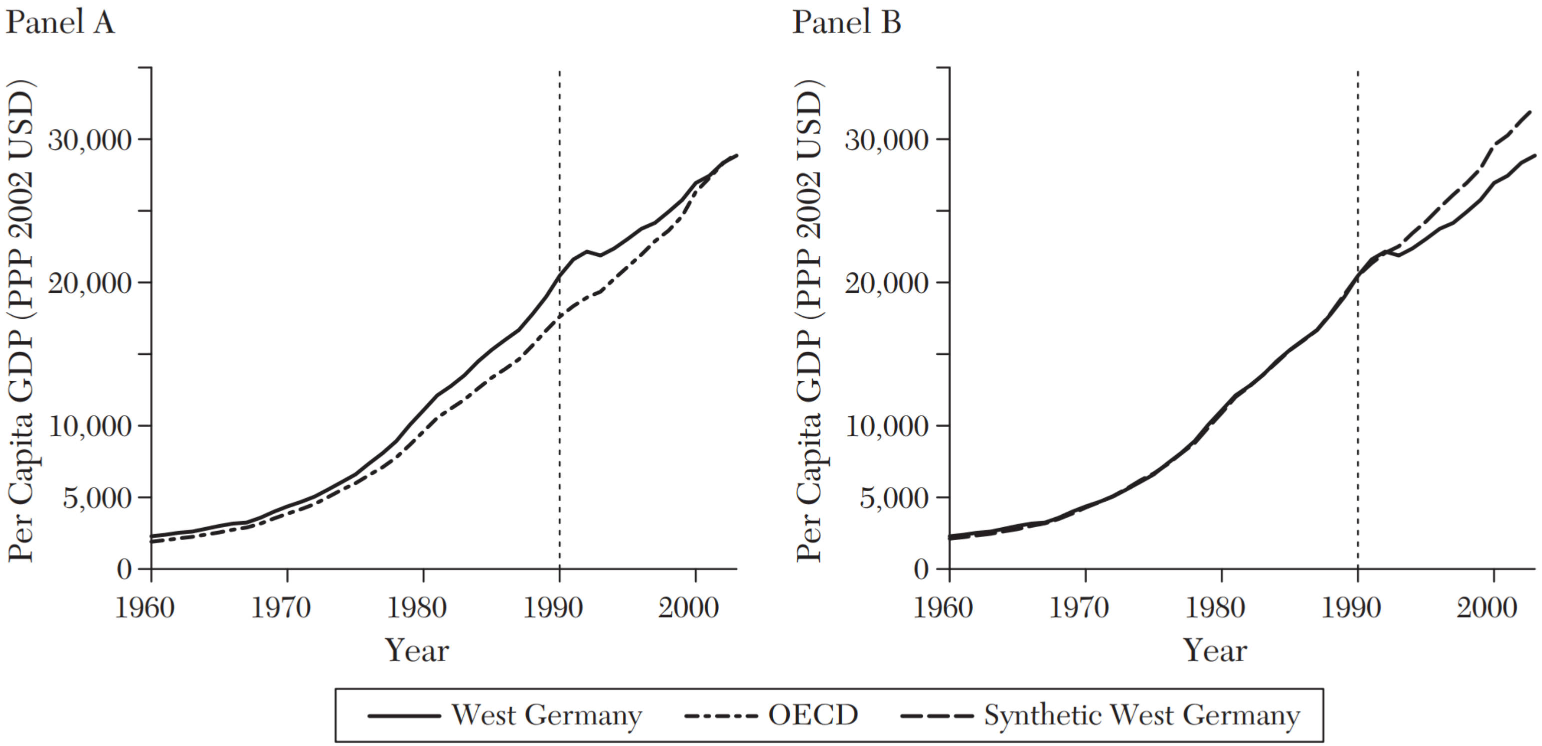


TABLE 2
SYNTHETIC CONTROL WEIGHTS FOR WEST GERMANY

Australia	—
Austria	0.42
Belgium	—
Denmark	—
France	—
Greece	—
Italy	—
Japan	0.16
Netherlands	0.09
New Zealand	—
Norway	—
Portugal	—
Spain	—
Switzerland	0.11
United Kingdom	—
United States	0.22

Donor Pool

Transparency of the counterfactual is one of the most attractive features of the synthetic control.

Abadie, A., 2021. Using synthetic controls: Feasibility, data requirements, and methodological aspects. *Journal of Economic Literature*, *59*(2), pp.391-425.

Programmable Potential Outcome: Synthetic Control

What if..? 인과추론의 데이터과학

- Synthetic control ≅ Predicted counterfactual

"The synthetic control approach developed by Abadie, Diamond, and Hainmueller (2010, 2014) and Abadie and Gardeazabal (2003) is arguably **the most important innovation in the policy evaluation literature in the last 15 years**. This method builds on difference-in-differences estimation, but uses systematically more attractive comparisons." (Athey and Imbens 2017, p. 9)

"Like for the lasso, **the goal of synthetic controls is out-of-sample prediction**" (Abadie 2021, p. 408)

Athey, S. and Imbens, G.W., 2017. The state of applied econometrics: Causality and policy evaluation. *Journal of Economic Perspectives*, *31*(2), pp.3-32.
Abadie, A., 2021. Using synthetic controls: Feasibility, data requirements, and methodological aspects. *Journal of Economic Literature*, *59*(2), pp.391-425.

Programmable Potential Outcome: Synthetic Control

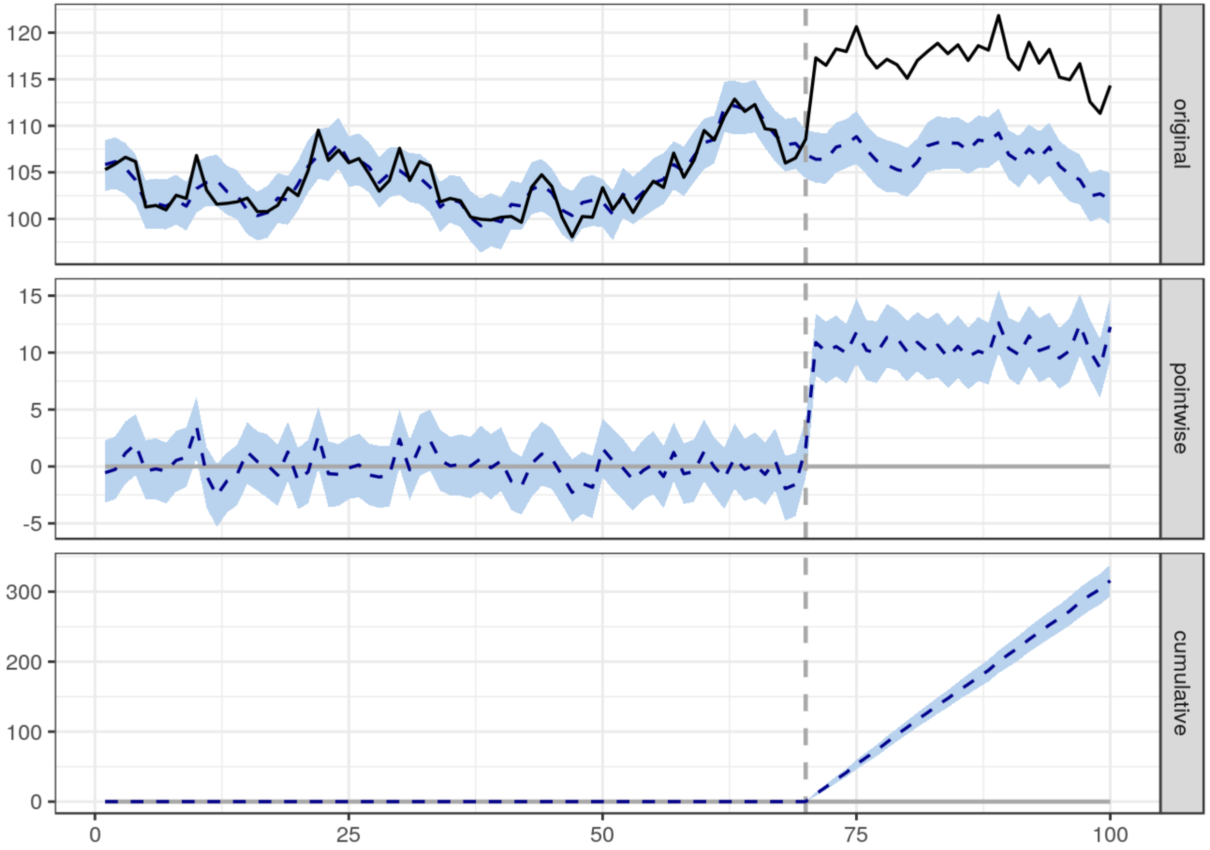
- Synthetic control ≅ Predicted counterfactual
  - Example: How Uber automated and scaled up the synthetic control

https://www.youtube.com/watch?v=j5DoJV5S2Ao

Programmable Potential Outcome: Interrupted Time-Series

What if..? 인과추론의 데이터과학

- Synthetic control ≅ Predicted counterfactual from the control group
- Interrupted time-series ≅ Predicted counterfactual from the prior trend
    - Example: Google’s CausalImpact (http://google.github.io/CausalImpact/CausalImpact.html)



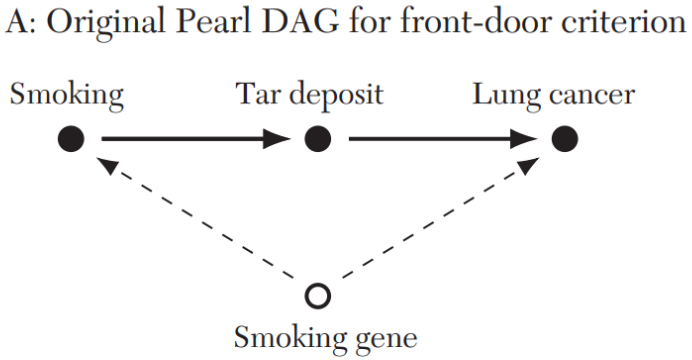
Brodersen, K.H., Gallusser, F., Koehler, J., Remy, N. and Scott, S.L., 2015. Inferring causal impact using Bayesian structural time-series models. *The Annals of Applied Statistics*, *9*(1), pp.247-274.

Limitation of Structural Causal Model: Unverifiable Graph

- Graphical models (Structural Causal Model) rely heavily on the *given* causal structure.

  'If you tell me how the world works (by giving me the full causal graph), I can tell you the answers on causal effects.'

- Example: Effect of smoking on lung cancer (Pearl and Mackenzie 2018; Imbens 2020)



Pearl, J. and Mackenzie, D., 2018. *The book of why: the new science of cause and effect*. Basic books.
Imbens, G.W., 2020. Potential outcome and directed acyclic graph approaches to causality: Relevance for empirical practice in economics. *Journal of Economic Literature*, 58(4), pp.1129-79.

Limitation of Structural Causal Model: Unverifiable Graph

What if..? 인과추론의 데이터과학

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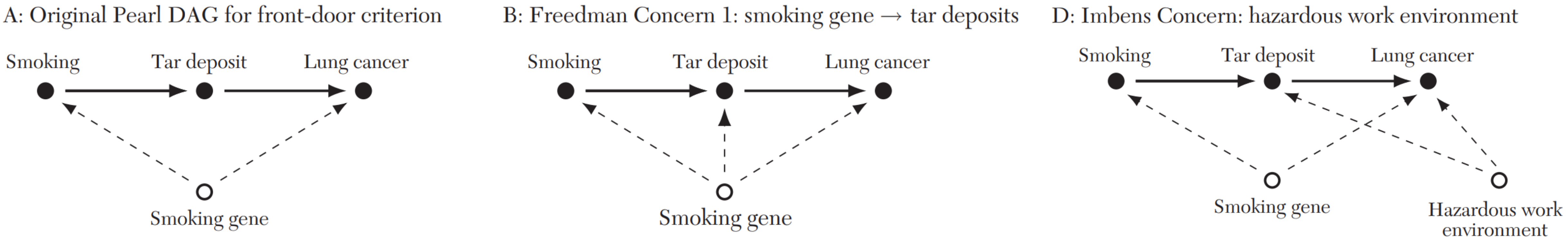


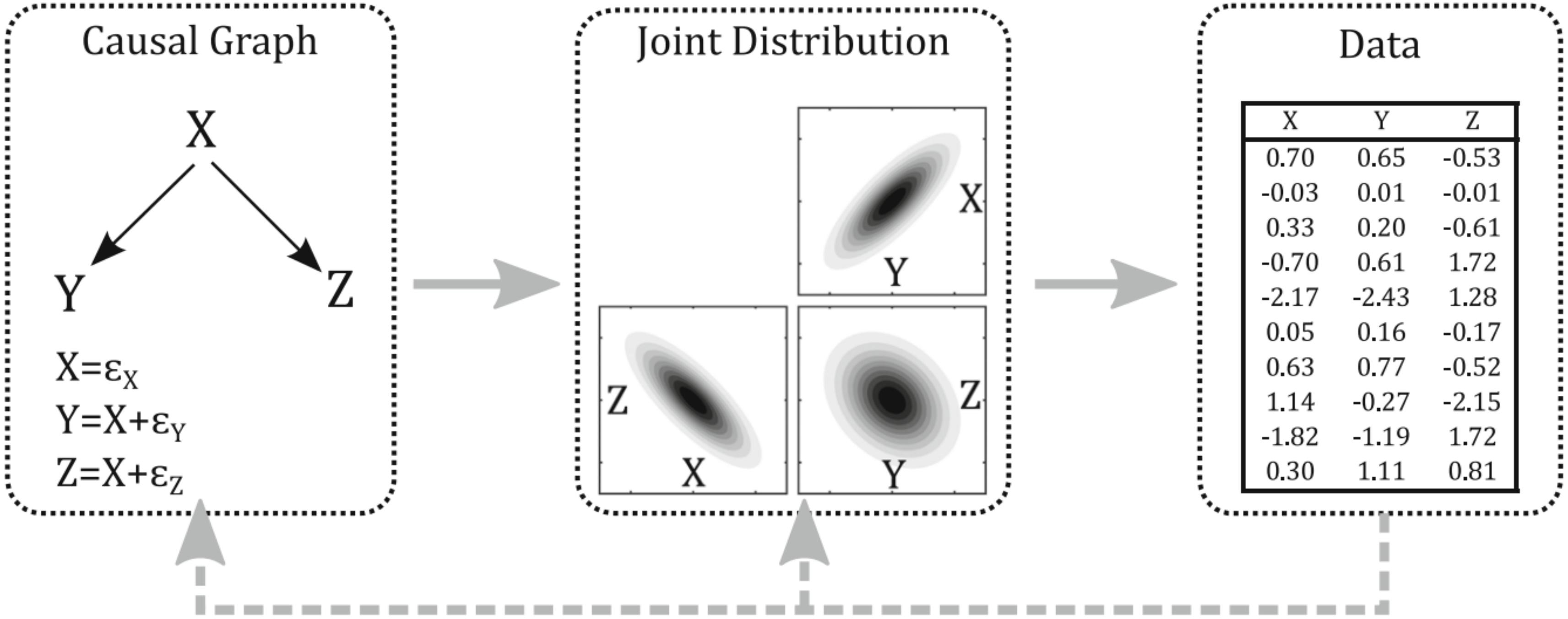
“*I have no quarrel with Freedman’s criticism in this particular example. I am not a cancer specialist, and I would always have to defer to the expert opinion no whether such a diagram represents the real-world processes accurately.*” (Pearl and Mackenzie 2018, p. 228)

Pearl, J. and Mackenzie, D., 2018. *The book of why: the new science of cause and effect*. Basic books.
Imbens, G.W., 2020. Potential outcome and directed acyclic graph approaches to causality: Relevance for empirical practice in economics. *Journal of Economic Literature*, 58(4), pp.1129-79.

Verifiable (Partial) Graph: Causal Discovery

- **Data Generation Process**: Causal Graph → Data
- **Causal Discovery**: Data → Causal Graph

Causal Effect Identification and Estimation



X	Y	Z
0.70	0.65	-0.53
-0.03	0.01	-0.01
0.33	0.20	-0.61
-0.70	0.61	1.72
-2.17	-2.43	1.28
0.05	0.16	-0.17
0.63	0.77	-0.52
1.14	-0.27	-2.15
-1.82	-1.19	1.72
0.30	1.11	0.81

Fig. 1 Relationships among the causal graph, joint distribution, and data. *Solid gray arrow* represents the data generation process, whereas the *dashed gray arrow* represents causal structural learning from data.

Ma, S. and Statnikov, A., 2017. Methods for computational causal discovery in biomedicine. *Behaviormetrika*, *44*(1), pp.165-191.

Limitation of ML/AI Models: Not Causal

"Some data scientists argue that the essence of intelligence is the ability to predict, and therefore that good predictive algorithms are a form of AI. From this point of view, large chunks of data science can be rebranded as AI… Rather, a hallmark of intelligence is the ability to predict counterfactually how the world would change under different actions by integrating expert knowledge and mapping algorithms. No AI will be worthy of the name without causal inference." (Hernán et al. 2019, p. 49)

"Machine learning does not represent an increase in artificial general intelligence [a.k.a. *strong AI*] of the kind that could substitute machines for all aspects of human cognition, but rather one particular aspect of intelligence: prediction." (Agrawal et al. 2019, p. 31)

"Explainable AI and interpretable machine learning have focused on helping researchers understand how algorithms work… A similar effort is now needed to help those who are not AI experts make sense of the output of algorithms and use them to guide decision-making." (Zheng et al. 2020, p. 21)

Hernán, M.A., Hsu, J. and Healy, B., 2019. A Second Chance to Get Causal Inference Right: A Classification of Data Science Tasks. *Chance*, *32*(1), pp.42-49.
Agrawal, A., Gans, J.S. and Goldfarb, A., 2019. Artificial intelligence: the ambiguous labor market impact of automating prediction. *Journal of Economic Perspectives*, 33(2), pp.31-50.
Zheng, M., Marsh, J.K., Nickerson, J.V. and Kleinberg, S., 2020. How causal information affects decisions. *Cognitive Research: Principles and Implications*, *5*(1), pp.1-24.

Causal ML/AI Models: Causal Representation Learning

What if..? 인과추론의 데이터과학

Toward Causal Representation Learning

This article reviews fundamental concepts of causal inference and relates them to crucial open problems of machine learning, including transfer learning and generalization, thereby assaying how causality can contribute to modern machine learning research.

By Bernhard Schölkopf, Francesco Locatello, Stefan Bauer, Nan Rosemary Ke, Nal Kalchbrenner, Anirudh Goyal, and Yoshua Bengio

ABSTRACT | The two fields of machine learning and graphical causality arose and are developed separately. However, there is, now, cross-pollination and increasing interest in both fields to benefit from the advances of the other. In this article, we review fundamental concepts of causal inference and relate them to crucial open problems of machine learning, including transfer and generalization, thereby assaying how causality can contribute to modern machine learning research. This also applies in the opposite direction: we note that most work in causality starts from the premise that the causal variables are given. A central problem for AI and causality is, thus, causal representation learning, that is, the discovery of high-level causal variables from low-level observations. Finally, we delineate some implications of causality for machine learning and propose key research areas at the intersection of both communities.

KEYWORDS | Artificial intelligence; causality; deep learning; representation learning.

Schölkopf, B., Locatello, F., Bauer, S., Ke, N.R., Kalchbrenner, N., Goyal, A. and Bengio, Y., 2021. Toward Causal Representation Learning. *Proceedings of the IEEE*, *109*(5), pp.612-634.

Talk by Yoshua Bengio on this topic:
https://www.youtube.com/watch?v=rKZJ0TJWvTk

Further Learning

- [Session 18-1] 가상의 통제집단 (Synthetic Control)
- [Session 18-2] 가상의 통제집단 분석 사례
- [Session 18-3] 데이터 기반의 인과관계 발견 (Causal Discovery)

Conclusion

Korea Summer Session on Causal Inference 2021

인과추론의 데이터과학

@ Zoom	Date & Time (KST) (1 ~ 1.5 hours)	Topic	Speaker
Module 1. Research Design for Causal Inference			
Session 1	July 6, 10 am (Tue)	Potential Outcomes Framework: Causal Mindset	Jiyong Park (UNC Greensboro)
Session 2	July 7, 10 am (Wed)	Overview of Research Design for Causal Inference	Jiyong Park (UNC Greensboro)
Session 3	July 9, 11 am (Fri)	Causal Inference in Business Research (1) Randomized Experiment	Hyeokkoo Eric Kwon (NTU), Nakyung Kyung (NUS)
Session 4	July 10, 10 am (Sat)	Causal Inference in Industry (1) A/B Test	Younseok Lee (ex-Coupang)
Session 5	July 12, 10 am (Mon)	Causal Inference in Business Research (2) Quasi-Experiment	Yongjin Park (CityU HK), Yoonseock Son (Notre Dame)
Session 6	July 14, 10 am (Wed)	Configurational Approach for Causal Recipes	YoungKi Park (George Washington)
Session 7	July 16, 10 am (Fri)	Graphical Model of Causal Relationships	Jiyong Park (UNC Greensboro)
Session 8	July 17, 10 am (Sat)	Causal Inference in Industry (2) Causal Diagram	Hojae Lee (NCSoft)
Session 9	July 21, 10 am (Wed)	Instrumental Variables	Jiyong Park (UNC Greensboro)
Session 10	July 23, 10 am (Fri)	Causal Inference in Biomedical Research: Meta-Analysis and Generalizability	Hwanhee Hong (Duke)
Module 2. Machine Learning for Causal Inference			
Session 11	July 26, 10 am (Mon)	Overview of Machine Learning for Causal Inference	Jiyong Park (UNC Greensboro)
Session 12	July 28, 10 am (Wed)	Applications of Machine Learning in Marketing	Jongho Kim (Cornell)
Session 13	July 30, 10 am (Fri)	Causal Applications of Machine Learning Models	Daehwan Ahn (UPenn)
Session 14	August 3, 11 am (Tue)	Heterogeneous Treatment Effect Estimation Using Machine Learning	Yejin Kim (UT Health)
Session 15	August 4, 10 am (Wed)	Causal Decision Making and Prescriptive Analytics	Jiyong Park (UNC Greensboro)
Session 16	August 6, 10 am (Fri)	Causal Inference under the Rubric of Structural Causal Model	Yonghan Jung (Purdue)
Session 17	August 9, 10 am (Mon)	Dataset Construction for Causal Analysis: Data Science for COVID-19 (DS4C)	Jimi Kim (UT Dallas)
Session 18	August 11, 10 am (Wed)	Synthetic Control / Causal Discovery	Jiyong Park (UNC Greensboro)

Design-Based Approach

Graph-Based Approach

ML for Causal Inference

Special Topics

Bridging the Two Worlds

Korea Summer Session on Causal Inference 2021

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Design-Based Approach

Lecture videos and materials can be found at

https://sites.google.com/view/causal-inference2021.

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Bridging the Two Worlds

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